

Combined Handouts of Environmental Psychology (PSY511)



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Lesson 1

Introduction to Environmental Psychology

The origins of environmental psychology can be traced back to the 1960s when researchers from various disciplines such as architecture, sociology, and psychology began to examine how the environment influences human behavior, cognition, and well-being (Steg, van den Berg, & de Groot, 2012). Initially, the focus was on the design and evaluation of built environments, such as homes, offices, schools, and hospitals. This was also known as architectural psychology or human factors (Clayton & Saunders, 2012). Later, the field expanded to include the natural environment, such as parks, forests, and wildlife. This was also called conservation or green psychology (Clayton & Saunders, 2012). Environmental psychology is a diverse and interdisciplinary field that addresses various topics, such as environmental perception, attitudes, and values, environmental behavior and decision-making, environmental stress and coping, environmental restoration and well-being, and environmental education and communication (Steg et al., 2012). Different environments can arouse different feelings in human beings. Our mental concepts and associations with a particular environmental type i.e., our perceptions about different surroundings also affect our physiology, feelings, and reactions. Any change in the environment also brings about a shift in our thinking styles and adaptive behaviors. Let's consider the following two situations:

1. Suppose you are sitting in your comfortable room with a cup of coffee while working on your laptop. Automatically you will feel pleasant and more motivated to involve yourself in creative tasks.
2. Now imagine that due to some emergency, you have to work on your laptop during a traffic jam while on public transport. You will feel unpleasant, your creative abilities will probably decline and you will think about getting rid of the task as soon as possible.

The first situation will perhaps give rise to feelings such as contentment, peace, joy, and relief. You will be able to think of multiple solutions to a single problem and develop a broadened understanding of the different demands that the environment presents. The second situation will probably give rise to feelings such as aggression, irritability, urgency, and dissatisfaction. It might even make you feel claustrophobic! In conclusion, it is fair to say that our environment and its different aspects can generate different feelings in us and ultimately different behavioral responses. With technology and rapid industrialization, the environment is becoming increasingly complex. The dangers of acid rain, the unreliability of nuclear power plants, and the difficulties in handling and disposing of toxic chemicals are being discussed. Additionally, the pace of suburban sprawl and urban decay is speeding up. Chemicals such as dioxin, formaldehyde, PCB (polychlorinated biphenyl), / and sulfur dioxide, once found only in the laboratories are becoming household words. Human beings are using and abusing the environment openly.

Many people are working hard to fix the environment, but there are still a lot of problems. Scientists, engineers, political and spiritual leaders, and the general public care more

about the environment now, but they don't always have the money, skills, or tools to do something about it.

The Environmental Debate

There is a debate regarding the environment between two groups of people. The first group has a pessimistic approach and the second group has an optimistic approach. Let's look at both:

► The Pessimists:

Some people say that the environment is so damaged that it will not be livable by the year 2100. They are angry about how many animals and plants have died in the last 100 years, how much land has become dry and useless, how many trees have been cut down, and how much water has been polluted by acid rain. They think that the environment is being harmed by the rich and the poor fighting over natural resources. They say that the rich take too much and the poor want to keep what they have, and this makes the environment suffer and lose.

► The Optimists:

Some people confidently highlight the remarkable achievements that have been made in improving human conditions and well-being in the fields of agriculture, medicine, and design technology. They believe that human creativity and innovation can overcome any challenge. They argue that there is sufficient energy, but the methods of obtaining it must be improved. There is a balanced population, but the distribution of people needs to be adjusted. There is manageable waste, but the systems of transporting and storing it need to be refined. They have a positive outlook on the future of the environment.

► Has the debate been resolved?

The debate on the environmental crisis remains unresolved, as different perspectives and solutions are proposed. Technology may offer some relief to the strain that human activities impose on the earth's resources, or human behavior may undergo significant changes to reverse the environmental damage. The intensity of the debate is increasing, as evidenced by the United Nations Earth Summit held in June 1992 in Rio de Janeiro, which raised global awareness of the situation. Along with this growing concern, there has been a renewed interest in the impact of the environment on human functioning. Various biological, psychological, and social problems have been predicted due to environmental degradation and population growth. Research is accumulating on the physiological, psychological, and behavioral effects of environmental factors such as noise, air pollution, temperature, barometric pressure, and building design. Both short-term and long-term effects are being examined. Moreover, some studies suggest that exposure to certain environmental conditions can have delayed negative consequences that persist even after the exposure has ceased. These are also being investigated with more attention. Likewise, the possibility of changing values, attitudes, and behavior toward the environment is being explored.

Definition of Environmental Psychology

To define Environmental Psychology, first, we need to understand the words

“Environment” and Psychology separately.

➤ **Environment: Mcqs and short question :**

Psychologist Kurt Lewin (1951) proposed a simple formula to define behavior (B) as a function of the person (P) and the environment (E)

$$B = f(P, E).$$

He stated that P and E are interdependent (ایک دوسرے پہ منحصر) (variables).

However, Wohlwill (1970) observed that the environment was often understood as social or interpersonal influences, or as an unclear concept. Saegert (1987) also criticized that the social sciences defined the environment mainly in terms of social interactions and structures. Thus, the term environment has been vague and ambiguous, referring to a general physical and social context for behavior. The term environment has also been applied to various nonsocial conditions, ranging from climate to bathroom design. All in all, the environment is the natural world that surrounds us and supports our life. It includes all the living and non-living things that affect us, such as animals, plants, water, air, land, and climate. The environment is important for our survival and well-being, as it provides us with resources, services, and beauty.

➤ **Psychology:**

Psychology is the scientific study of the mind and behavior. Psychology often focuses on understanding a person's emotions, personality, and mind through scientific research, experiments, observation, and trials. Psychology often involves **four goals**, including the following:

- **Describe:** This goal involves collecting information and naming, classifying, and summarizing what is observed. For example, a psychologist may describe the symptoms of depression, the stages of memory, or the types of personality.
- **Explain:** This goal involves understanding the causes and reasons behind the observed behavior. For example, a psychologist may explain why people are more likely to help others in some situations than others, how stress affects the immune system, or what factors influence intelligence.
- **Predict:** This goal involves forecasting when and how the observed behavior will occur again in the future. For example, a psychologist may predict how a person will perform on a test, how a couple will resolve a conflict, or how a group will make a decision.
- **Change:** This goal involves modifying, influencing, or controlling the observed behavior to achieve positive outcomes. For example, a psychologist may change a person's habits, attitudes, or emotions, help a client cope with a problem, or design an intervention to prevent or treat a disorder.

Psychology is a science because it uses the scientific method to study and understand human behavior and mental processes. The scientific method is a systematic and objective way of collecting, analyzing, and interpreting data based on empirical evidence

and logical reasoning. The scientific method involves four steps: (1) making observations and asking questions, (2) forming hypotheses and making predictions, (3) testing hypotheses and gathering evidence, and (4) drawing conclusions and communicating results. Psychologists use the scientific method to test theories, discover facts, and solve problems in various areas of psychology, such as cognitive, social, developmental, and clinical psychology. By using the scientific method, psychology aims to produce valid, reliable, and generalizable knowledge that can be applied to real-world situations (Cherry, 2020; McLeod, 2020).

➤ **The relationship between humans and the environment:**

Humans and the environment have a complex and interdependent relationship, as they influence and affect each other in various ways. Humans depend on the environment for their survival and well-being, as it provides them with natural resources, such as water, food, energy, and materials. However, humans also modify and transform the environment for their benefit, often causing harm and degradation to it. For example, humans cause climate change, pollution, deforestation, biodiversity loss, and environmental injustice through their activities and consumption patterns (Clayton & Myers, 2015). These environmental problems have negative impacts on human health, quality of life, and security, as well as on the ecosystems and species that share the planet with humans (Steg, van den Berg, & de Groot, 2012).

Therefore, humans need to find ways to improve their relationship with the environment and make it a win-win situation for both. This means that humans should respect and protect the environment, and use it sustainably and ethically, so that does not compromise its ability to support life and function. This also means that humans should benefit from the environment, and enjoy its beauty, diversity, and services, without harming or exploiting it. A win-win relationship between humans and the environment can be achieved by various means, such as adopting green lifestyles, technologies, and policies, promoting environmental education and awareness, engaging in environmental activism and stewardship, and supporting environmental justice and equity (Clayton & Myers, 2015; Steg et al., 2012).

➤ **So how can Environmental Psychology be defined?**

- The branch of psychology which deals with the scientific study of relationship between physical environment and human behavior is called environment psychology.

Environmental psychology is an interdisciplinary field that studies the complex and dynamic relationships between humans and their environments. It draws from various disciplines, such as biology, geology, psychology, law, geography, economics, sociology, chemistry, physics, history, philosophy, and engineering. It aims to understand how humans and environments affect and interact with each other. It also focuses on solving socially relevant problems and applying knowledge in practical ways. It considers both the influence of the physical environment on human behavior and the influence of human behavior on the physical environment.

Environmental psychology could, therefore, be defined as a behavioral science that investigates, to enhance, the interrelationships between the physical environment and human behavior.

This definition does not show how rich and different environmental psychology is, or its goal to make better ways of understanding. Environmental psychologists want to know how people's bodies, minds, and actions react to their environment. Researchers in the field, therefore investigate questions that involve physiological content (e.g., changes in heart rate, endocrine functioning, galvanic skin response, mortality)", psychological content (e.g., spatial behavior patterns, mental images, environmental stress, attitude change) behavioral content (e.g. altruism, aggression, performance). They also adopt an interdisciplinary perspective, and consider the meteorological, physical, geographical, architectural, and ecological features of the environment that affect its inhabitants. These aspects are also relevant to other disciplines, such as geology, physics, chemistry, and biology. However, environmental psychology tries to integrate these aspects and understand the interrelated processes that govern organism-environment relationships. Moreover, environmental psychologists balance between solving practical problems and developing broad-based integrative theory. They use theoretical and applied research to inform each other. For instance, they use theory to understand and prevent negative reactions to crowding and use practice to refine theories of crowding. Therefore, environmental psychology is a unique and diverse discipline that draws from various disciplines, methods, concerns, and theories. Considering these factors, and the possibility of change and criticism, we will define environmental psychology as:

A multi-disciplinary behavioral science, both basic and applied in orientation, whose foci are the systematic interrelationships between the physical and social environments and individual human behavior and experience.

Goals of Environmental Psychology:

The goals of environmental psychology are to:

- Understand how and why our environment impacts us, such as our behavior, emotions, and cognition.
- Apply that knowledge to improve the design and management of environments that promote well-being, productivity, and sustainability, such as green buildings, parks, and public spaces.
- Improve our relationship with the world around us, by fostering environmental awareness, conservation, and social responsibility

Objectives of the course:

- To learn about the history, scope, and methods of environmental psychology
- To understand how human behavior and the environment influence and interact with each other • To explore the interdisciplinary nature of environmental psychology and its connections with other fields
- To develop an environmental perspective in psychology and apply it to various issues and problems
- To examine the theoretical and empirical developments in environmental psychology

- and their implications
- To anticipate the future trends and challenges in environmental psychology and their solutions
 - To analyze the environmental challenges faced by Pakistan and their psychological aspects

Common Assumptions (عام مفروضے) of Environmental Psychology:

The term environment is a vague and broad concept that can be interpreted differently by various disciplines. For instance, historians, biologists, architects, sociologists, economists, and psychologists have each adopted their definitions of the environment to suit their research interests and objectives. Historians have explored the environmental Zeitgeist, biologists have examined the ecological niche, architects have analyzed the design features and economists have calculated the ratio of supply to demand to explain how external factors influence human behavior. Despite the ambiguity of the term environment, it is possible to identify some common assumptions that underpin its study in all environmental science, regardless of the specific approach. These assumptions are as follows:

- (1) The earth is the only suitable habitat we have.
- (2) The earth's resources are limited.
- (3) The earth as a planet has been and continues to be profoundly affected by life.
- (4) The effects of land use by humans tend to be cumulative.
- (5) Sustained life on earth is a characteristic of ecosystems and not of individual organisms or populations.

We will begin our next lesson by discussing these assumptions in detail.

Lesson 2

Common Assumptions of Environmental Psychology

The term environment is a vague and broad concept that can be interpreted differently by various disciplines. For instance, historians, biologists, architects, sociologists, economists, and psychologists have each adopted their definitions of the environment to suit their research interests and objectives. Historians have explored the environmental Zeitgeist, biologists have examined the ecological niche, architects have analyzed the design features and economists have calculated the ratio of supply to demand to explain how external factors influence human behavior. Despite the ambiguity of the term environment, it is possible to identify some common assumptions that underpin its study in all environmental science, regardless of the specific approach. These assumptions are as follows:

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- (3) The earth as a planet has been and continues to be profoundly affected by life.
- (4) The effects of land use by humans tend to be cumulative.

(5) Sustained life on earth is a characteristic of ecosystems and not of individual organisms or populations.

In the following pages, we will elaborate on each of these assumptions.

1. The earth is the only suitable habitat we have, and its resources are limited. It has been proven by research that human beings will also be affected if certain species become extinct. The earth has witnessed various forms of life emerge, evolve, flourish, and go extinct, with humans achieving a dominant position only in the recent past. However, this domination does not disprove two essential realities that must be acknowledged:

(1) *Humans, too, will succumb to extinction either through geologic, meteorological, or interstellar cataclysm, natural biological processes, internecine quarrels, or because the earth's resources will no longer support human life in its present form*

(2) *Although there may be other habitable islands in the cosmic sea, they are spaced at such great distances throughout the universe as to be virtually irrelevant to human survival.*

Despite these realities, the earth is of great significance to us and our future generations. We need to adapt to the opportunities and constraints that the earth offers, as well as the inevitable fact that it is constantly evolving. Moreover, we need to do so in a manner that ensures our survival. Environmental psychology, a new and emerging field, promises to provide insights that will enable us to live on earth in a sustainable way.

2. The earth as a planet has been profoundly (deeply) affected by life. (We need a lot of resources available here on earth to live, and if we don't have them, life will end." Earth supports our life by providing us with food, shelter, air, and water. Since all these resources are used by countless species, they are limited.")

Human contributions to the visual landscape of the planet are everywhere: tall skyscrapers, intricate networks of highways and electrical lines/ engineered lakes/ and the vapor trails of high-flying jet planes are constant reminders of human presence. More subtle indicators are the changing quality of the chemical composition of the atmosphere, geologic changes in the earth's crust, engineering changes in hydrologic processes and chemical changes in the waters that cover the planet.

Clearing of forests, plowing of land, black-topping of highways and parking lots not only affect the earth visually but change the amount of carbon dioxide in the atmosphere, the light reflecting and absorbing characteristics of the earth's surface and hydrologic drainage patterns. These changes in turn influence the rate of heating and cooling and therefore the temperature of the earth and its atmosphere. Weather and climate are thus influenced by the presence of humans. It is hoped that a more thorough understanding of the ways in which humans influence earth's natural processes will lead to more intelligent and life-preserving behaviors on their part.

3. The outcomes of land use by humans tend to be cumulative and therefore we have obligations to ourselves as well as future generations to minimize their negative effects.

انسانوں کی طرف سے زمین کے استعمال کے نتائج مجموعی ہوتے ہیں اور اس وجہ سے ہم اپنے اور آنے والی کے لیے ان کے منفی اثرات کو کم کرنے کے لیے ذمہ داریاں عائد کرتے ہیں) (نسلوں

Reading : A number of changes in the environment are brought about by human habitation. For example, fall plowing, the development of sanitation systems and the building of shelters. These practices influence the earth immediately as well as having long-term impact. A dam built today not only supplies electrical energy for today's population but also influences the course of the waterways on which the dams are built, often in irreversible fashion. The conclusion of all this is that while humans have multiplied, their life's resources have shrunk and there is every indication that this trend is continuing. For example, the same conditions that helped to create the Sahara are expanding it southward. Every year two to three more square miles are lost to the drought and sands. The "Thar" Desert of India is advancing at the rate of about one-half mile annually along its entire perimeter. We have an obligation to ourselves and to future generations to see that this trend is halted, if not reversed. An understanding of environmental psychology represents a start toward reversing these trends.

3. **Sustained (برقرار) life on earth is a characteristic of eco system and not of individual organisms and populations.** No single organism, population or species is capable of both producing all its own food and completely recycling all of its own metabolic products. **Green plants and light produce carbon dioxide, sugar and water;** from sugar and inorganic compounds many organic compounds are manufactured including protein and woody tissue. But no plant can degrade the woody tissue developed in this fashion back to its inorganic compounds. This degradation requires other organisms such as bacteria and fungi. To complete the recycling of chemical elements from inorganic to organic and back to inorganic requires the use of several species. **The smallest system capable of complete chemical recycling is known as ECO System.** Humans are part of a very complex and delicately balanced ecosystem. Thus, **to understand humans fully it is necessary to study them within the context of the ecosystems in which they survive.** Environmental psychology is the field that attempts to develop an understanding of this **interdependency.** **(All species are interdependent in eco system).** It should be clear by now that to comprehend fully the relationship of humans to their environment the student of the environment should be aware of contributions from a large number of disciplines. Environmental studies by their very nature are the domain of a generalist with a strong interdisciplinary interest. All environmental problems must be looked at from numerous perspectives so that a clear and total picture can be put together from the many pieces.

After studying the common assumptions of **Environmental Psychology** in detail, it is important to state the comprehensive definition of Environmental Psychology once again. Students must focus how these common assumptions are tightly packed in the following definition:

A multi-disciplinary behavioral science, both basic and applied in orientation, whose foci are the systematic interrelationships between the physical and social environments and individual human behavior and experience.

This definition mentions that Environmental Psychology is "Basic" and "Applied" in orientation.

Let's briefly look at these two terms:

- **Basic Behavioral Science.**

Basic behavioral science is a term that refers to the scientific study of human behavior and its underlying causes and consequences. Basic behavioral science aims to discover general laws and theories that explain and predict human behavior, as well as to identify the factors that influence and modify it. An example of basic behavioral science is the field of behavioral psychology, which studies how people learn, remember, and behave based on their interactions with the environment. • **Applied Behavioral Science.**

Applied behavioral science is the use of scientific knowledge and methods from various behavioral science disciplines, such as psychology, sociology, anthropology, and economics, to solve practical problems or create new products or technologies that involve human behavior. An example of applied behavioral science is the field of applied behavior analysis (ABA), which involves applying behavioral principles and techniques in various settings, such as education, health care, criminal justice, and social services, to help individuals with behavioral or developmental challenges.

Current Events Influencing Environmental Psychology

If present trends continue: The world will become increasingly crowded, more polluted (Leonard, 1986; Peterson, 1987), less stable ecologically (Manabe & Wetherald, 1986), and more vulnerable to disruption (World Commission on Environment and Development, 1987). Supplies of drinking water will diminish drastically, and despite greater material output, the world's people will become poorer than they are today (Postel, 1985, 1986, 1987). The forests of the world will become increasingly shed as a result of the requirements of wood for building and burning (Bowander, 1987; Myers, 1984) and as a result of the increasing acidity of rainfall worldwide. The need for commercial land will increase and less land will be left behind for farming and cultivation. The following factors are important to be considered in this regard.

- **Population Trends.**
- **Resource Depletion and Environmental Degradation.**
- **Public Policy and the Environment.**
- **Human Behavior and the Environment.**

1. Population trends

According to the world population projections, if the current trends continue, the human population will reach 10 billion by 2030 and 30 billion by the end of the twenty-first century. These numbers are close to the estimated maximum number of people that the earth can support. However, some regions of the world have already surpassed the limit of their carrying capacity. The population in Sub-Saharan Africa and the Himalayan regions of Asia has exceeded the ability of their local environment to sustain life. The combination of overgrazing, fuel wood collection, and harmful farming practices has resulted in a series of ecological changes from open woodlands to scrub to fragile semi-arid land to useless weeds and finally to bare soil. The situation is worsened by the burning of animal dung and crop residues for heating and cooking, which deprives the croplands of organic matter, reduces their water retention, and lowers their fertility. In Bangladesh, Pakistan, and large parts of India, the high demand for basic needs by the

large population has degraded the quality of the cropland, pasture, forests, and water resources that their livelihood relies on. Let's look at a few population statistics in **Pakistan.**

Population mid-2007	169,300,000
Births per 1000 population	31
Deaths per 1000 population	8
Rate of natural increase (percent)	2.3 %
Projected Population, 2025	228,800,000
Projected Population, 2050	295,000,000

The dramatic increase in world population has been a driving force for the development of Environmental Psychology.

In this lecture, only "Population Trends" was discussed in detail. The remaining three current events will be discussed in detail in the next lecture.

Lesson 3

Theories in Environmental Psychology

We will begin lesson 3 by discussing the following current events influencing Environmental Psychology.

- Population Trends. (already discussed in detail in the lecture 2)
- Resource Depletion and Environmental Degradation.
- Public Policy and the Environment.
- Human Behavior and the Environment.

2. Resource Depletion and Environmental Degradation.

Other examples of serious deterioration in the earth's basic resources can be found throughout the world, including the industrialized nations. Erosion of agricultural soil, salinization of the highly productive irrigated farmland, and Lake Acidification are becoming more prevalent. Extensive loss of tropical forests and more or less permanent soil degradation have occurred throughout much of the Amazon River Basin (McHale, 1979). The world's deserts now make up 800 million hectares increasing in size.

Regional water shortages and deteriorating water quality which is already a problem in many parts of the world, are likely to worsen. The world water withdrawals are expected to increase by 200 to 300 percent in the next two decades, and a large proportion of it will be polluted due to the use of waterways as coolants and as waste carriers. The potential for human conflict over the use of freshwater reservoirs is heightened by the fact that out of the 200 major river basins of the world, 148 are shared by two countries and 52 are shared by three to ten countries. Long-standing difficulties over the shared rivers of Plata (Brazil and Argentina), Euphrates (Syria and Iraq), and Ganges (Bangladesh and India) have the potential to intensify as the need for additional freshwater occurs. Thus, the depletion of natural resources essential for survival has been another factor influencing the development of environmental psychology.

The industrialization, urbanization, and population growth of the planet have resulted in various negative consequences. Chemical and human waste is being produced at rates faster than we can safely dispose of, non-renewable resources are being consumed at increasing rates, plans for reestablishing renewable resources are shortsighted and the fate of our waterways, wildlife, climate, and perhaps the earth itself appears in jeopardy. The deterioration of the environment is thus another reason for the increased interest in environmental psychology.

Let's look at a few facts about population trends in Pakistan.

Pakistan population trends and environmental indicators

CO ₂ emissions per capita in 2002	0.7 metric tons
Natural habitat remaining (%)	63 %

*We have consumed 37% of our natural habitat we are left with.

3. Public Policy and the Environment

(public policy is an instrument which regulates the behavior of people in specific community.) If the trees are being cut, they should also be replanted with the same amount. Policies should be formulated to maintain or increase the productive capacity of land.

Some possible policies to achieve this goal include:

- Sustainable agriculture practices
- Soil conservation and reclamation
- Efficient water management
- Land-use planning and management
- Climate-smart agriculture)

The problems in preserving the carrying capacity of the earth and sustaining the possibility of a decent life for human beings are indeed enormous and about to happen but they are just that— problems. Policy changes coupled with government business and individual actions can do much to alleviate many of them. Reforestation after cutting requires detoxification of chemical byproducts before disposal as well as judicious soil management, thereof, the government has started to implement policies in this regard. Interest in energy and material conservation is growing, industrial and household recycling is becoming more prevalent and the need for family planning is becoming better understood. High-yield hybrids are continuously being introduced and methods for farming the seas are being developed. Public policy requires dependable scientific evidence to support its decisions. This has created a pressing need for the advancement of environmental psychology and has fostered its expansion as a field of study. However, inspiring change in people with regard to environmental literacy such as encouraging them to use personal bags for shopping rather than black plastic bags also requires economic resources. This is because we would have to provide an alternative to the black plastic bag business when our efforts will pull down their industry. This calls for a well-rounded environmental policy to be developed by the government and environmentalists combined.

4. Human Behavior and the Environment

(This includes behavior change, how we can change human behavior to benefit the environment.)

Here are some strategies to promote behavior change:

- 1: Education and awareness:
- 2: Social influence: Encourage social norms that support sustainable behaviors.
- 3: Community engagement: Organize local events, campaigns, and initiatives that foster a sense of community and environmental responsibility.

Although these policy developments are positive steps, they are insufficient to address the increasing difficulties that humanity faces. Needed changes go far beyond the responsibility of one nation, and new initiatives must be taken if worsening poverty, environmental degradation, and international conflict are to be prevented. The solutions if they exist are complex and long-term and tied to the problems of poverty, injustice, and social conflict. The problems of the globe are human problems, many of which have been caused directly or indirectly by human presence. Advanced technology, while

potentially a part of the solution to some of these problems, is also the cause of many of them. Because many of these problems are the result of human behavior, the psychologist, whose domain of interest is human behavior, has potentially a great deal to contribute in their resolution.

Environmental psychologists are very much attuned to the health of mother earth. They realize there are no quick fixes and that only through understanding and changing human behavior can there be any hope of maintaining a habitable planet—that is, even though psychologists recognize that many of the problems are technological they also emphasize that the source of these problems is human behavior. Equally important is their realization that without an understanding of the mechanisms and laws that govern the life-sustaining processes of the earth, there is no hope that any changes in human behaviors or policies will have any medicinal effects.

Psychologists in this regard tend to focus on behavior modification. An example of behavior modification with reference to environmental psychology is the use of token economy to promote pro-environmental behaviors, such as recycling, saving energy, or using public transportation. A token economy is a system in which individuals earn tokens or points that can be exchanged for rewards. For instance, a school may implement a token economy to reward students for bringing reusable water bottles, turning off lights, or taking the bus. The tokens can then be used to buy items from the school store, such as pencils, stickers, or books. This way, the students are motivated to adopt more eco-friendly habits.

Theories in Environmental Psychology

Environmental psychology is an interdisciplinary field that benefits from the diverse viewpoints of various disciplines on a common phenomenon, which may result in less biased theories and more universal solutions. However, this diversity also poses challenges for the consistent use of knowledge and the formulation of coherent and comprehensive theories.

Let's begin by discussing four theories that come under the umbrella of Historical Influences.

Historical Influences

Thinking among environmental psychologists has been influenced by theories both within and outside of the discipline of psychology. Some of these theories are very: (i) Broad in scope:

The scope of a theory means the kinds of problems that the theory addresses. Theories that are broad in scope try to understand numerous phenomena all at once. Their natural framework is so vast and so general in its nature that it can accommodate a bulk of information. (ii) Focused in scope:

Theories with a narrow or focused scope specifically try to understand a particular thing. For example, a specialist in cell biology might conduct research on cells only while a generalist in biology might engage in surface-level research on a wide range of topics. (iii) Empirical:

These are scientific in nature and are data-driven. For example, the theory of Natural

selection by Charles Darwin which is supported by a lot of data driven research. (iv)

Non-Empirical:

These are general theories usually not backed up by data-driven research. They are based on logic and intuition. For example, those proposed by Plato in ancient philosophy.

We will review a number of them to provide the context for a consideration of current theories of environment-behavior relationships. **These theories include:**

- **Geographical Determinism**
- **Ecological Biology (theories)**
- **Behaviorism (behaviorist perspective or interactionism)**
- **Gestalt psychology**

Geographical Determinism

Some historians and some geographers have attempted to account for the rise and fall of entire civilizations on the basis of environmental characteristics. For example, Toynbee (1962) theorized that the environment (specifically, topography, climate, vegetation availability of water etc)presents challenges to its inhabitants (باشنده) . Extreme environmental challenge leads to the destruction (تباہی) of a civilization, whereas too little challenge leads to stagnation (جمود) of culture. Thus, Toynbee proposed that an intermediate level of environmental challenge enhances the development of civilizations, and extremely diminished or excessive levels are debilitating. The notion of environmental challenge and behavioral response, although rooted in the thinking of such geographical determinists, appears often in one form or another in various theories in environmental psychology. As one example Barry, Child, and Bacon (1959) have suggested that agricultural, non-nomadic cultures seem to emphasize responsibility, obedience, and compliance in child rearing practices, whereas nomadic cultures often emphasize independence and resourcefulness. These differences, they suggest, result from the fact that people who live and work together in organized non-mobile communities require more structured organization and therefore stress the importance of obedience and compliance. On the other hand, independence and resourcefulness are esteemed and taught by nomads in preparation to meet the changing and unpredictable demands of an environment confronted by them. Thus, the environment sets the stage for the development of cultures having the best chance of survival.

This implies that ghetto culture develops a set of skills in its residents that are most appropriate for the ghetto. It can be reasoned that someone who is not proficient in street fighting is unlikely to cope with the conditions of the streets. As we will discuss later in this text, some have extended this argument to other established environments that shape their residents in ways that are seemingly adaptive to those environments but are harmful in the broader context.

Ecological Biology (theories)

Environmental Psychology has been greatly influenced by the development of ecological theories, which examine the biological and sociological interrelations between

organisms and their environment. Ecological science has shown that organisms are not isolated from their environment, but rather essential to it. This idea of organism-environment mutualism is present in many contemporary environment-behavior theories. **The environment and its inhabitants are often treated as distinct elements, but their interdependence is undeniable.** These various elements are considered to form a holistic system and **any change in one element is expected to affect the whole system.**

We will discuss Behaviorism (behaviorist perspective or interactionism) in detail in the next lecture.

Lesson 4

Arousal Theories

We start this lecture in continuation of the last lecture by discussing Behaviorism (behaviorist perspective or interactionism) that comes under the influences of Historical Influences.

➤ **Behaviorism (behaviorist perspective or interactionism)**

It comes from the psychology discipline and involves behaviorists' reaction to the failure of personality theories to account fully for human behavior. It is now generally accepted that considering both the environmental context in which behavior occurs and person variables (i.e., personality, dispositions, attitudes, etc.) leads to more accurate behavior predictions than measurement alone. This is central to most current theories of environmental behavior. Human beings are biological entities that share this environment along with other species. For example, Human beings and elephants have a long and complex history of coexistence, ranging from cultural reverence and mutual benefit to conflict and competition. (King et al., 2016). Human behavioral patterns are characterized by the attitude that they have. For example, if we consider why people use black plastic shopping bags despite knowing their negative consequences, we will conclude that this has to do with their personal preferences and attitudes. Top behaviorist of the 20th century, William James also said:

"The greatest discovery of my generation is that human beings can alter تبدیل their lives by altering the way they look at things"

Let's try to understand the crisscross between environmental and personal factors in predicting human behavior by looking at the biography of Muhammad Ali Jinnah.

Muhammad Ali Jinnah was born in Karachi on December 25, 1876. He was admitted to bar in England and then he returned to India and became an active supporter of The Indian National Congress. In 1913, he joined the Muslim League. In 1916, he was elected the president of the Muslim League, he was re-elected to this post again in 1920. He played an important role in bringing about the Lucknow Pact of 1916. In the Lucknow Pact, the Congress agreed to ensure a separate electorate for the Muslim community. He resigned from Congress in 1930 and channeled all his efforts towards the ends of

the Muslim League. He headed the Muslim League and struggled until the creation of the Independent Pakistan. He was appointed the first Governor general of Pakistan.

After reading this biography, ask yourself one question: Did Muhammad Ali Jinnah's personal characteristics make him the founder of Pakistan, or was it the result of environmental factors? This question will help you completely understand the role of heredity and environment in shaping human behavior.

➤ Gestalt Psychology

Gestalt psychology was developed primarily in Germany and concurrently with behaviorism. Gestalt psychologists were more concerned with perception and cognition than with overt behavior. Perception is how we make sense of our surroundings, cognition is how the information coming from the environment is processed in our brain, and overt behavior is how we react in response to our information processing. For example, a child might tell his mother that he wants to be a fighter pilot and start watching pilot movies and cartoons because of the kind of information he received and processed about fighter pilots. In the same manner, an employer might reject a very talented candidate for the job just due to his negative perception of him. The most important principle of this body of work was the objects, persons, and settings are perceived as a whole, which is greater than the sum of its parts. From the Gestalt point of view behavior is rooted in cognitive processes; it is determined not by stimuli, but from the perception of those stimuli. For example, a man might react positively to a high traffic jam because of his positive mood due to his recent promotion, and a man who recently fought with his wife, due to his negative mood, might become angry due to even a minor delay caused by the traffic.

Mini-Theories

These approaches (the historical influences) tend to be rather broad in scope and lacking in empirical evidence. Each has its focus on convenience and no single perspective is satisfactory in accounting for the complexity of the environment-behavior relationship. No "grand theory" exists that can incorporate the distinctive contributions of each of these influences on environmental psychology. This is so for at least four reasons:

- (1) There is not enough data available regarding environment-behavior relationships to lead to the kind of confidence needed for a unifying theory.
- (2) The relationships that researchers have looked at are highly varied.
- (3) The methods used are inconsistent.
- (4) How variables have been measured has not always been compatible from one research setting to the next.

Even though well-articulated, all-encompassing theories are not available at this time, there are several "mini-theories," or mini-approaches, that have been used successfully in conceptualizing specific organism-environment relationships. Included in these are the arousal approach, the stimulus load approach, the behavioral constraint approach, the adaptation level approach, the environmental stress approach, and the ecological approach. Each of these can handle some, but not all, of the available

data. Some are more useful in dealing with group behavior (the ecological approach), whereas others focus on the individual level of analysis (the stimulus load approach). Some find their greatest utility at the psycho-physiological level (the stress approach); others are useful for accounting for individual differences (the adaptation level approach). Before turning to an orientation that attempts simultaneously to embrace aspects of many of these approaches, a brief description of each of these approaches will be provided.

➤ Arousal Theories

Arousal theories have typically been concerned with the influence of arousal on performance. Generally, performance is maximized at intermediate levels of arousal but falls off as arousal is either increased or decreased. (Arousal is a state in which someone feels excited or alert).

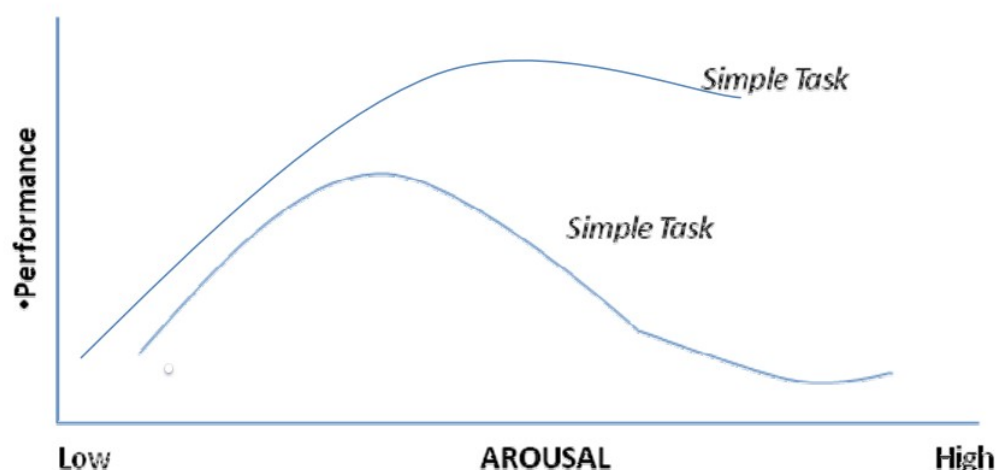
If the arousal is at the highest or lowest level, the performance will not be good. When the arousal is at a medium level, the performance will be good. This relationship, sometimes referred to as an inverted-U relationship, has been shown to differ slightly depending on whether performance is measured on simple or complex tasks (see graph Figure) and is often referred to as the Yerkes-Dodson law. These relationships are consistent with other findings that humans tend to seek out intermediate levels of stimulation (Berlyne, 1974), and are reminiscent of Toynbee's assertion, referred to earlier that cultures only develop in environments that provide intermediate environmental challenges. Example :

A student who becomes careless about his papers may not perform well because his arousal level is low.

If a student gets too worried about his papers, he may not perform well because his arousal level is too high.

We should learn to maintain arousal only then we can give good performance.

Performance is predicted to be optimal for both simple and complex tasks at intermediate levels of arousal. Arousal above the level leads to decrements in performance



Performance changes that vary curvilinear with temperature increases have also been shown. One explanation of these findings is that increases in ambient temperature lead to increases in arousal levels. Initially, the higher arousal leads to performance enhancement, but as it increases further, over-arousal occurs, causing performance decrements. Similarly, it has been shown that personal space invasions lead to increases in arousal (Middlemist, Knowles, & Matter, 1976) and performance decrements (Evans & Howard, 1972; McBride, King, & James; 1965). Additionally, increases in noise levels have been associated with changes in arousal and performance (see Evans & Cohen, 1987). Thus, several variables associated with arousal changes are related to performance changes, and performance has consistently been curvilinearly related to arousal. Other theorists utilizing an arousal perspective have featured physiological responses to environmental stimulation. Changes in heart rate, blood pressure, respiration rate, galvanic skin response, and adrenaline secretion among others have been shown to occur with changes in the environment. Increased ambient temperature leads to blood vessel dilation, perspiration, increased heart rate, and in extreme conditions, lowered blood pressure and insufficient oxygen supplied to the brain. Personal space invasion has been linked to delayed onset and shorter duration of maturation for males. Noise exposure alters blood pressure, heart rhythm, and the flow of gastric juices to the stomach.

Neurologists such as Hebb (1972) have linked arousal with increased activity of the reticular activating system of the brain. Still, other theorists have equated arousal with changes in motor activity or with self-report of arousal. Berlyne (1974) has, for example, characterized arousal as lying on a continuum anchored at one end by sleep and at the other by excitement, and Mehrabian and Russell (1974) have identified arousal as a major component in people's affective responses to their environment.

Independent of the orientation taken concerning arousal, several consistencies are apparent:

1. Changes in arousal are associated with changes in the environment
2. Pleasant as well as unpleasant stimulation increases arousal – that is, room temperatures above 100 degrees Fahrenheit and loud, obnoxious noises influence arousal in ways similar to rollercoaster rides.
3. Changes in arousal lead people to seek information about their internal states (Schacter & Singer, 1962) as well as to seek information from others (Festinger, 1954)
4. People tend to evaluate moderate levels of arousal positively
5. Often great expenditures of energy are utilized by individuals to bring the environment to a level of moderate stimulation.

► Factors Impacting Arousal Level

Furthering the point various researchers have tried to identify the impact of different environmental factors on the arousal level:

- I. **Temperature:** An increase in ambient temperatures leads to an increase in arousal level. Usually, higher arousal leads to performance enhancement but as it

increases further over-arousal occurs causing performance decrease.

II. **Personal Space Invasion**: An optimal personal space invasion can increase performance while too much invasion can cause a decrease in performance. For this reason, cubicles are designed in offices in such a way that colleagues are clustered together yet have a separate working space. This provides an optimal invasion of personal space leading to increased arousal (due to collaboration, relative performance assessment, and comparison) which ultimately leads to increased performance.

III. **Noise Level**: An optimal noise level can increase arousal and performance while too much noise can decrease arousal and performance. For example, you might be able to work better while sitting in a room while occasionally chatting with your colleagues (due to optimal noise level) and you might not be able to work appropriately in a room where there is too much chatter and loud music because noise level is not optimal.

➤ **Physiological Response to Environmental Stimulation**

Changes in heart rate, blood pressure, respiration rate, galvanic skin response, and adrenaline secretions have been shown to occur with changes in the environment. Every human carry 12 British thermal units of electricity, which can create fluctuations in the body's ecosystem/metabolism or equilibrium when it comes into contact with environmental stimuli.

I. **Temperature**

An increase in ambient temperature leads to the following physiological changes:

- Blood vessel dilation • Pupil dilation • perspiration • Increased heart rate.

II. **Extreme conditions**

Extreme environmental conditions such as food, water, or nutritional scarcity can lead to:

- Lowered blood pressure • Insufficient oxygen reaching the brain due to which a person might collapse.

II. **Invasion in personal space has been linked to**

- Delayed onset and shorter duration of micturition for males.

II. **Exposure to Noise:**

Extreme noise may:

- Alters blood pressure, heart rhythm, and flow of gastric juices to the stomach which might lead to acidity.

➤ **Arousal and Nervous System:**

Arousal is linked with increased activity of the reticular activating system of the brain.

- ✓ For arousal theorists, consistency and break inconsistencies تضادات are also very important. For example, in a park, we might not notice people engaging in exercise and laughter because that's what everyone is doing but if we see a man crying in the same park, it will be something unusual and inconsistent with what was happening in the environment. It will immediately grab our attention and increase our arousal and ultimately behavior.

- ✓ **Changes in arousal are associated with changes in the environment.** Pleasant as well as unpleasant stimulation increases arousal, either negative or positive. i.e. room temperatures above 100 degrees might stimulate negative arousal, while a 25-degree temperature may stimulate optimal arousal.
- ✓ **Changes in arousal lead people to seek information about their internal states.** For example, during times of high arousal and stress, people might experience hot and cold body flashes. This might make them want to inquire about their blood pressure or sugar levels.
- ✓ **People tend to associate moderate levels of arousal positively.** This is why the interior of many offices is designed to be calm and cool so that people experience positive arousal and ultimately perform better at work. Great expenditure of energy is utilized by individuals to bring such specific environments to a moderate level of stimulation.

Lesson 5

Stimulus Load, Behavioral Constraint, and Adaptive Level Theories

► Stimulus Load Theories

Central to stimulus load theories is the notion that humans have a limited capacity to process information. When inputs exceed that capacity, people tend to ignore some inputs and devote **وقف کرنا** more attention to others (Cohen, 1978). These theories account for responses to environmental stimulation in terms of the organism's momentary capacity to attend to and deal with salient features of its environment. Generally, **stimuli most important to the task at hand are allocated as much attention as needed and less important stimuli are ignored**. Our attention is basically like a switch. We can control our attentional capacity according to the demands that we face from the environment. The benefit of this quality of attention is that humans can meet the changing demands of the environment where whereas the downside of this quality of attention is that it becomes limited when environmental demands are too high.

Example

For example, while driving during rush-hour traffic a great deal of attention is paid to the cars, trucks, buses, and road signs around us, and less attention is paid to the commentator on the car radio, the kids in the back seat, and the clouds in the sky. If the less important stimuli tend to interfere with the task at hand, then ignoring them will enhance performance, (e.g., ignoring the children's fighting will make you a better and safer rush hour driver. If, however, the less important stimuli are important to the task at hand, then performance will not be optimal; for example, ignoring the road signs because you are attending to the more important trucks, cars, etc., may lead you thirty miles out of your way in getting home (Figure below).

While driving during rush hour traffic a great deal of attention is paid to the cars, trucks, buses, and road signs and less attention is paid to the commuter or the car radio, the kids in the back seat, and the clouds in the sky.



Sometimes the organism's capacity to deal with the environment is overtaxed or even depleted. When this occurs only the most important information is attended to, with all other information filtered out. Once attentional capacities have been depleted even small demands for attention can be draining. Thus, behavioral aftereffects including errors in judgment, decreased tolerance for frustration, ignoring others in need of help, and the like, can be accounted for by these theories. For example, the exhausted rush hour driver eventually might reach the point where he or she doesn't notice the traffic

light turn from red to green (or worse yet, from green to yellow to red), even though this is a very important stimulus. Additionally, decreased tolerance for frustration may lead to "Laying on the horn" or "lane hopping" and motorists in the break-down lane may be ignored, if not looked upon with dislike.

Read the following explanation given by an athlete about how he tries to limit the focus of his attention on the track and you will understand how attention functions in the real world.

ATTENTION IN THE REAL WORLD

"I have learned to cut out all unnecessary thoughts... on the track. I simply concentrate. I concentrate on the tangible - on the track, on the race, on the blocks, on the things I have to do. The crowd fades away and the other athletes disappear and now it's just me and this one lane" Stimulus load theories are also able to account for behavioral effects in stimulus-deprived environments (e.g., certain behaviors occurring aboard submarines and in prisons). That is, this approach suggests that understimulation can be just as aversive as overstimulation. So-called cabin fever resulting from monotonous living conditions can also be seen as the result of understimulation. Wohlwill (1966) has argued that environments should be depicted in terms of measurements applied to the dimensions of intensity, novelty, complexity, temporal variation, surprisingness, and incongruity, all of which contribute to stimulus load. Subsequent behaviors can then be related to the stimulus properties of environments in systematic and comparable ways. Let's look at these dimensions briefly one by one.

(i) Intensity:

It means the range of a stimulus for example if the music is loud it means that it is of high intensity and if the music is of less volume it means that it is of low intensity. We will experience more stimulus load if the music is high and versatile.

(ii) Novelty:

It is the uniqueness of a stimulus. For example, if there are 100 channels on the TV, often, the most unique one will grab our attention and contribute to stimulus load. (iii) Complexity:

It is how complicated or multi-faceted a stimulus is. For example, driving through rush hour. Multiple cars, noises, and other traffic-related stimuli come into play and contribute to stimulus load.

(iv) Temporal Variation:

This is specifically related to changes in noise level. For example, imagine that you are sitting on a bus, and the AC of that bus is making a repetitive sound. In the beginning, that voice might irritate you but slowly you will get used to it and will no longer be able to notice it as it will get mixed up with other sounds. But suddenly the bus undergoes a jerk and the AC voice gets fixed on its own, then you will automatically feel that something is missing from the environment and you will finally notice the change. These noise variations also contribute to stimulus load. (v) Surprisingness:

This is a stimulus that is quite rare so it invokes پکارتا ہے the emotion of surprise in a person. For example, imagine that you come to know that a new restaurant has opened in town and that restaurant is made up of ice and chocolate. You will be extremely surprised and feel motivated to visit that restaurant. Such rare stimuli also contribute to stimulus load.

(vi) Incongruity ہم آہنگی:

This is a stimulus that holds together a unique pairing of two different stimuli that normally would not exist together. For example, if you see a three-year-old baby sitting in a cage with a lion, you will feel strange and question yourself how this strange situation came into being. Such unique pairings of stimuli also contribute to stimulus load.

Stimulus to Behavior-A relationship

After studying these stimulus characteristics, you might be wondering what is the relationship between stimuli and behavior. This automatically brings us to the implications of stimulus load theories. Since stimuli impact human behavior, we have a lot of control that we can exercise over human behavior by altering the stimuli around us. For example, offices and schools can be designed in such a way that they foster positive and productive behaviors.

► **Behavior Constraint Theories**

Behavior constraint theories focus on the real, or perceived, limitations imposed on the organism by the environment. An example of a real threat would be fear of failing in exams while an example of a perceived threat is a student feels that he will definitely fail, even after all the preparation. According to these theories, the environment can prevent, interfere with, or limit the behaviors of its inhabitants (Rodin & Baum, 1978; Stokols, 1978). Humans and the environment can control each other. For example, a student who is traveling on the train for the first time might feel that he is in danger. However, he might try to negotiate with the environment by making some friends or reading the travel guide for ease and certainty. Friday afternoon rush-hour traffic interferes with rapid commuting; loud, intermittent noises limit effective communication (for example a teacher might not be able to talk to students properly if there is too much noise in the classroom.); overregimentation (patients being rotated in far-off departments in hospitals for collaborative treatment) in hospitals can interfere with recovery, excessively high ambient temperatures prevent extreme physical exertion, and extremely cold temperatures limit finger handiness. In a sense, these theories deal with situations where persons either lose some degree of control over their environment, or they perceive that they have.

Brehm and Brehm (1981) assert that when we feel that we have lost control over the environment, we first experience discomfort and then attempt to reassert دوبارہ قائم کرنے our control. They label this phenomenon as **Psychological Reactance**. (which is the reaction of the person to regain the lost control over the environment. For example, a teacher might first feel discomfort in a noisy class but then she will try to apply different classroom strategies to control the noise level). If the rushhour traffic interferes with getting home in a timely fashion, we may leave work early, or find alternate, less-congested routes. Loud, intermittent noises may be dealt with by removing their source or by changing environments. Extreme temperatures are handled by adjusting the thermostat. All that is needed is for individuals to perceive that they have lost some degree of control, or for that matter, to anticipate the loss of control, and reactance will occur. If repeated attempts to regain control are unsuccessful, **learned helplessness** may develop (Seligman, 1975). People begin to feel as though their

behavior does not affect the environment. They begin to believe they no longer control their destiny, and that what happens to them is out of their control. These feelings can eventually lead to clinical depression, and in the most extreme form can lead people to give up on life, and die.

On the opposite side of the coin, perceived control over one's environment (even when real control does not exist, or is not used) can alleviate the negative outcomes that the environment might otherwise bring about. Perceived control over noise (Glass & Singer, 1972), overcrowding (Langer & Saegert, 1977), and over one's daily affairs (Langer & Rodin, 1976) has been shown to influence positively a variety of behavioral responses. For example, residents of a nursing home who were given greater control over, and responsibility for their well-being displayed enhanced mood and greater activity in comparison with residents who were not given control. Similarly, people who had control over the thermostats in their working and living environments reported fewer health complaints during the winter months than those who did not have control. These results occurred even though they did not manipulate the thermostats and kept their environments at ambient temperatures similar to those without control (Veitch, 1976). Behavior constraint theories thus emphasize those factors (physical as well as psychological; real as well as imaginary) associated with the environment that limits human action.

► Adaptation-Level Theories

Adaptation theories are similar to stimulus load theories in that an intermediate level of stimulation is postulated to optimize behavior. Excessive stimulation as well as too little stimulation is hypothesized to have deleterious effects on emotions and behaviors. For example, a dull home interior might make a person feel depressed and a highly stimulating interior might make a person restless. An optimal and calm home environment will make people feel comfortable. Major proponents of this position include Helson (1964) and Wohlwill (1974). While all environmental psychologists emphasize the interrelationship of humans to their environment, adaptation-level theorists speak specifically of two processes that make up this relationship—the processes of adaptation and adjustment. Organisms either adapt (i.e., change their response to the environment) or they adjust, (i.e., change the environment with which they are interacting). For example, if political figures are in prison and are deprived of the number of social interactions that they used to have in their daily life, they might adapt and adjust by calming down and focusing their energies on reading and writing about their political journeys. Adaptations to decreases in ambient temperature include piloerection (hair on the body standing up or what is commonly called getting "goose pimples"), muscle rigidity, increased motor activity, and vasoconstriction; adjustments include throwing another log on the fire or turning up the thermostat. Either process brings the organism back to equilibrium with its environment.

This perspective values the recognition of personal adaptation levels, meaning how much excitement or activity a person is used to and looks for in their surroundings. This idea helps us understand why two people might react differently to the same situation. For instance, someone who enjoys lots of excitement might find a noisy party fun, but it

could be too much for someone who likes things calmer. In the same way, while some might enjoy the busy feeling of shopping right before Christmas, others might not like it when the store gets too crowded. These personal preferences for excitement can lead to very different choices. A person who likes a lot of excitement might go to lively parties, while someone who doesn't might stay away or find a quiet spot if they do go. We've all noticed people who are the center of attention at a party and those who prefer to keep to themselves. The way they act can often be linked to their comfort with different levels of excitement.

Lesson 6

Ecological Theories and Comparison of Theories

We start this lecture by discussing environmental stress theories in detail

Environmental Stress Theories

Stress theories emphasize the mediating role of physiology, emotion, and cognition in the organism-environment interaction. Environmental features are seen as influencing, through the senses, the organism, causing a stress response when environmental features exceed some optimal level. For example, initially, a guitar is at rest. But when you start to play the guitar, your finger movement causes stress in the strings. This can be imagined as a disturbance in the homeostasis level (or in other words, the "rest" level) of the guitar. The organism then responds in such a way as to alleviate the stress. **For example**, before feeling hungry, your body is steady (homeostasis is balanced). When you feel hungry the homeostasis of your body is disturbed (i.e., release of gastric juices in the stomach, body weakness). To alleviate this stress, you look for food. After eating the food again your homeostasis comes back to normal. Part of the stress response is automatic. Initially, there is an alarm reaction to the stressor, wherein various physiological processes are altered. Resistance then follows as the organism actively attempts to cope with the stressor. Finally, as coping resources are depleted, a state of exhaustion sets in (Selye, 1956).

Types of stress:

There are two main types of stress:

1. Eustress:

It is a positive force that provides a sense of well-being. This stress is up to an ideal degree. **For example**, a girl is participating in a college debate. She feels very anxious and starts imagining/visualizing again and again that she has to go in front of people. This image disturbs her homeostasis. Her physiology, emotions, and cognitions are disturbed and she enters a state of an unbalanced mind. Suddenly she starts to think of strategies through which she can well prepare for her public speaking assignment. It is Eustress that motivates her to prepare for her task. After good preparation and a successful debate her unbalanced mind again becomes balanced and homeostasis is restored.

2. Distress:

It is a negative force that can lead to a decline in well-being. This stress is up to an unsafe degree. For example, the girl in the above example goes to the college debate and faints بے ہوش in front of people, runs away from the hall, or starts experiencing a panic attack in the hall. It is distress that leads to her failure in the college debate.

Let's look at these reactions to stress proposed by Hans Selye (1907-1982) in greater detail:

1. **Alarm Reaction:** This is the first reaction to stress when an organism first acknowledges or perceives stress in the environment. It is the mobilization of resources to confront the stress. For example, a child is happily playing in the park. Suddenly he sees a dog. His first reaction (Alarm reaction) to this stress will be that he will start feeling nervous. This nervousness will prepare his body to deal with the stress and his physiological/mental resources will come into play.
2. **Resistance:** This is the stage where an individual decides his strategy for dealing with the stress. The threat to the environment will lead to more energy consumption. To acquire more energy the physiological reserves of energy will be used and energy consumption will keep on increasing according to the demands of the stressful situation. For example, a man suddenly hears a loud sound. He comes to know that a lion has escaped his cage and sees it coming after him. His resistance reaction will be that he will start to run while continuously consuming his energy by sweating, breathing heavily, muscle movement, or consumption of body glucose.
3. **Exhaustion:** Now imagine that the lion follows the man for 5 hours. This will lead him towards the exhaustion stage. Exhaustion is when stress continues for long periods. Energy is used but it is not refilled. A perfect example would be war/combat between two countries that has continued for 2 years. All the soldiers have constantly faced life-threatening situations for two years and they are being exposed to extreme stress. There is a shortage of food, manpower, and ammunition. So, their resources are not re-filled. At one point they will experience exhaustion or you can say that they will enter into survival mode.

How do we know that we are experiencing stress?

Stress has two common indicators:

1. **Physiological:** Physiological indicators of stress are critical markers for assessing the body's response to stressors. According to a review article, common physiological indicators include changes in blood pressure, heart rate, and levels of blood fats such as cholesterol and triglycerides (Iqbal et al., 2021). Additionally, variations in blood sugar levels, particularly in the evening, and alterations in appetite, which can contribute to weight gain, are also noted as physiological responses to stress (The Journal of Clinical Endocrinology &

Metabolism, 2021).

2. **Emotional:** Emotional indicators of stress are key to understanding how individuals react to stressors. Research has shown that common emotional responses to stress include feelings of irritability, anxiety, and depression (Katana et al., 2019). These emotional states can significantly impact daily functioning and are associated with various mental health outcomes (Sarrionandia et al., 2018).

Ecological Theories

Central to the thinking of ecological theorists (Barker, 1963, 1968) is the notion of organism environment fit. An example of a career would be perfect here. A person with excellent analytical, organizational, and mathematical skills will be fit for the job of a banker or a software engineer. Another example would be the selection of a partner for marriage. Two people courting each other might come to know that they are a perfect fit for each other because they have similar values, outlooks, and understanding in life. Environments are designed or grow to accommodate certain behaviors. Behavior settings, as Barker termed them, are evaluated in terms of the goodness of fit between the interdependent environmental features and the behaviors that take place. For example, a schoolyard, a church, a classroom, an office, or an entire business organization might be considered a behavioral setting; each would then be evaluated in terms of how suitable it is for the play behavior of children, how well it accommodates the religious rituals, or how well it serves the functions of the business.

While any number of behaviors can occur within any physical setting, cultural purpose is defined by the interdependency between standing patterns of behavior and the physical setting. Standing patterns of behavior represent the collective behavior of the group rather than just individual behavior. The standing pattern of behavior in a classroom would include lecturing, listening, observing, sitting, taking notes, asking questions, and taking tests; the physical environment of this behavior setting would include the room and such accessories as a lectern, chairs, chalkboard, microphone, overhead projector, and slide screen. Because this standing pattern of behavior occurs primarily in this behavior setting, social ecologists would suggest that knowing about the setting helps us predict what will occur in it. Once individuals making up this classroom leave this physical setting, most of the physical characteristics of the environment remain unchanged, but the behaviors likely will change dramatically in other words, the students will move to a new physical environment eliciting a different standing pattern of behavior.

Examples:

1. The initial concept kept in mind when designing prisons is that this is a place where people are confined as a punishment for crimes. Prisons are also built away from housing and commercial buildings so that criminals are kept away from general people to prevent them from harm. However, a lot can change if we

start looking at prisons in a different manner. Prisons must not only be considered as “punishment houses” but also as “correctional facilities”. Criminals are people with psychological problems due to which they commit different crimes. It was us who had a wrong perception of them. They are not to be hated. This viewpoint about prisons and prisoners may lead us to change the total structure of prisons. Physical settings can be designed in such a way that different standing patterns of behaviors can be produced in them. Hospitals can be built inside prisons to provide prisoners with complete care. Prisons can organize regular psycho-education lectures, exercise sessions, career guidance, and counseling as well as on-prison working opportunities for prisoners. All this might help them become better citizens after they are released from prison.

2. An environment conducive to agitation and crime will only lead to agitation and crime. Imagine a crowded traffic square of a South Asian country where there is a lot of noise, smoke, chaos, and traffic. Ask yourself. In this environment, there will be more crime or less crime/ more agitation or less agitation. There will be more agitation which will lead to increased frustration and increased chances of crime.

We will continue with additional details regarding Ecological Theories in the next lecture.

Lesson 7

The present framework and future directions in Environmental Psychology (I)

We will begin this lesson by continuing the topic of Ecological theories

Ecological Theories: undermanning (understaffing)/ overmanning (overstaffing)

According to Barker and Gump (see Gump, 1987), if the number of individuals in a setting falls below some minimum, some or all of the inhabitants must take on more than their share of roles if the behavior setting is to be maintained. **This condition is termed understaffing.** In an environmental context, understaffing is when there are too many resources/duties in the environment and a lesser number of organisms. For example, the college roommate of one of your authors went to a very small high school, which in many ways exemplifies an understaffed setting. This roommate played football and, like most high schoolers, played before fans on Friday night. Also, like most high schools, there was a high school band that performed at half-time. Your author's roommate, however, was also the best trumpet player in the school; so, at half-time, while the rest of the football players were obtaining instructions as to what they should be doing the second half, he was out marching with the band. In understaffed settings, inhabitants often have to assume a variety of roles.

If the number of participants in a setting exceeds, the capacity for that setting, **then the setting is considered overstaffed.** In an environmental context, overstaffing is when there are fewer resources and too many organisms. More or less the population in a geographical environment disturbs the ecosystem. It causes a positive or a negative

impact on the standing pattern of behavior. Too many swimmers waiting to get into the pool on a hot summer afternoon, commuters on the five o'clock train, football fans at the Super Bowl, and shoppers in department stores at Christmas time might all represent overstaffed settings.

Comparison of Theories

Theories	Approach
Arousal	General
Stimulus Load	Specific
Adaptation Level	Specific
Stress Approach	General
Behavioral Constraint	Specific
Ecological Approach	General

**** General in scope:** The scope of a theory means the kinds of problems that the theory addresses. Theories that are broad in scope try to understand numerous phenomena all at once. Their natural framework is so vast and so general that it can accommodate a bulk of information. For example, generalists in biology might engage in surface-level research on a wide range of topics.

**** Specific in scope:** Theories with a narrow or focused scope specifically try to understand a particular thing. For example, a specialist in cell biology might conduct research on cells only.

In the comparison of theories, we discuss what is the area of prediction of these theories. Whatever these theories address they can predict something about those behavioral patterns. We can use this predictive ability to exert control over our environment. Now let's look at the key statements of these theories and their areas of prediction.

Theory	Key Statement	Area of prediction
The Arousal Approach	The arousal approach is the most general in identifying physiological and affective mediators of environment behavior relationships	Increases or decreases in stimulation produce corresponding changes in physiological and psychological arousal and, in turn, produce predictable variations in behaviors such as task performance and aggression.

The Stimulus Load Approach	The stimulus load approach is synaptic, focusing on cognitive information-processing abilities.	Yields predictions regarding the social/behavioral consequences of over/under-stimulation—excessive attentional demands have differential effects on the performance of primary versus secondary tasks and the likelihood of attending to various social stimuli.
The adaptation level approaches	An intermediate level of stimulation is postulated to optimize behavior. Excessive stimulation as well as too little stimulation is hypothesized to have harmful effects on emotions and behaviors	Predicts consequences of a particular stimulation level depend on the specific level to which a given individual has adapted.
The Stress approach	The stress approach incorporates elements of all of the above. That is, stress can be characterized in terms of objective physical and social environmental conditions that deviate from some optimal level (e.g., a noisy and crowded subway) and are thus potentially disruptive to human functioning.	The approach has the further advantage of predicting behavioral coping and the consequences of adaptive and maladaptive coping mechanisms.
The Behavior Constraint Approach	The behavioral constraint approach is the most limited of all in scope, (i.e., it is primarily useful in situations where the	When such conditions do exist, the concepts of reactance and learned helplessness yield useful predictions of behavioral responses to such conditions.
	perception of loss of control or threats to control are present).	

The Ecological Approach	The ecological approach has the broadest scope with the concept of behavior setting, and as such is a useful descriptive approach to understanding the behaviors of large numbers of people in different settings.	How behavior settings impact behavior. however, this generality limits the approach's ability to account for individual differences in the behavior setting.
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How these theories help in real life: The case of Amir

After a comparative analysis of these theories let's look at one real-life example to understand how they may be applied in real life.

Let's suppose that there is a young man named Amir. He is 20 years old and recently got admission to a city university. He then moves to the city, leaving behind his village. Upon his arrival, he discovers that the university is very big. The city is also very large and commercialized, with heavy traffic jams, large shopping malls, parks, and open markets, and is densely populated. He finds out that there is a lot of distance between his residence and the university. To meet his living expenses, he also has to work a part-time job in the evening. He wakes up early in the morning, does breakfast at a local café, and takes on different buses to reach his university. After a hectic schedule of classes, he then again takes different buses to reach his workplace in the evening. After his work, he again takes different buses to reach back home where he barely finds time to have dinner. He then has to study and complete his assignments. After which he is left with very little time to sleep. The next day this routine repeats over and over again. Amir finds the entire environment very confusing, however, he keeps going on with the routine due to his determination towards his studies. But after six months Amir starts to experience headaches, nausea, and vertigo. He starts to dream of his village where he lived a happy life and felt close and connected to people and nature around him. He then decides to move back to his village.

After reading the case of Amir, keep the following questions in your mind:

1. How would you assess Amir's life circumstances? Happy or Stressful?
2. What features of his physical and social environment influence the quality of his life? Is it a positive influence or a negative influence?
3. What might you infer about his personality and thought processes that may interact with the environment to influence his experiences? Is it a positive inference or a negative inference?
4. Can you apply the concepts of arousal, stimulus load, behavior constraint, adaptation level, and behavior setting to account for Amir's thoughts, feelings, and behaviors?

Just think of these questions from the perspective of an Environmental Psychologist and try to understand how the theories of environmental psychology form the basis of

all research surrounding it.

The present framework and future directions in Environmental Psychology

- **The main purpose of the present framework:** The present framework for understanding organism-environment relationships.
- **Origin:** It stems from several theoretical positions and databases established within the domains of general and social psychology, as well as from the newly emerging area of environmental psychology. It is developed from theories emphasizing the affective components of the human experience, where overt behaviors, characterized as approach or avoidance responses, are seen as being mediated by the emotion-arousing properties of the environment.
- **Target:** To bring different theories and databases under the umbrella of a single model.
- **Specific focus:** Specifically, environmental factors (e.g., density, personal space, noise, temperature, etc.) will be postulated to influence individual affective states, which in turn will be asserted to influence overt behavior. For example, how much rise in ambient temperature gives how much arousal.

Following are seven postulates relevant to the present and future framework of Environmental Psychology.

First postulate:

- A. The environment consists of both physical and social variables existing in reciprocal relationships with behavior.
- B. The physical and social environments can and should be characterized in terms of measurements applied to their salient characteristics.

Second postulate:

- A. The environment, physical as well as social, typically exerts a steady-state influence on the behavior of its inhabitants.
- B. When the measured values of the environment's most salient characteristics undergo dramatic change, the environment's influence on behavior can no longer be characterized as a steady state.
- C. Disruptions of steady state occur when present perceptions do not correspond to the desired or expected level of social and physical stimulation.

Third postulate:

The influence exerted by the environment is indirect i.e. Environment acts to influence people's emotional states, which in turn mediate their overt behavior.

Fourth postulate:

Environment-evoked emotions are best depicted in terms of three distinct dimensions: pleasure – displeasure, degree of arousal, dominance –submissiveness.

Fifth postulate:

The environment–influenced behavior of the individual is dependent upon the extent and the configuration of the dimensions of emotions aroused.

Sixth postulate:

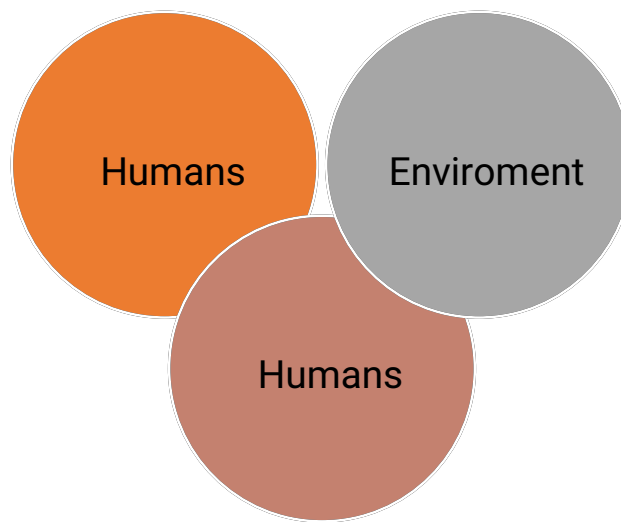
Behavior is goal-directed. That is to say behavior in an environmental setting is performed to increase or decrease the pleasure, degree of arousal, or dominance elicited by changes in the environment in an attempt to return the environmental influence to a steady state.

Seventh postulate:

When the goal-directed behavior is ineffective or when great expenditures of energy are required to maintain a steady state, the environment can be considered pathological and a disruptive influence on human functioning. **Let's now look at all of these seven postulates one by one in detail.**

First postulate:

1A: The environment consists of both physical and social variables existing in reciprocal relationships with behavior. Attempts to understand human/environment relationships that have focused exclusively on either the human components or the environmental aspects have not met with great success. **Humans are not only influenced by their environment, but through their behavior, they also alter that environment. The altered environment in turn produces subtle changes in the environmental inhabitant and its behaviors; these behaviors again produce subtle changes in the environment, and so on, indefinitely.** For example, the classroom behaviors of teachers and students alike are influenced by such physical properties of the environment as room temperature, chalkboard space, windows (or the lack thereof), the arrangement of desks, available light, and ambient noise level. However, the total classroom environment also includes social and demographic variables (e.g., the age and gender of classmates, the friendship groups, the type of activity being engaged in, and others). Each one of these factors, singly and collectively, influences the behavior that is enacted in this setting. By the same token, **the resultant behavior changes the nature of that setting.** For example, to enhance discussion, chairs may be moved, thus changing the environment; these changes may lead to changes in friendship groups or to a change in the ambient noise level. Teachers may now have to talk louder, perhaps also changing the tone of their voices, making them appear grumpy; this perception may in turn lead students to avoid the space around the teacher or may keep them from asking questions. Thus, the environment and its inhabitants never stay the same; each is constantly changing as a result of its interactions with the other. Any attempt to understand these relationships, therefore, requires a systematic approach emphasizing the bidirectional, often symbiotic, nature of this interaction.



1B. The physical environment can and should be characterized in terms of measurement applied to its salient characteristics. It is an obvious scientific advantage to be able to specify relationships between or among variables in precise mathematical terms. The symbolic representation of the effect x has on y is not as potentially ambiguous if x and y have properties that can be measured by agreed-upon and reliable techniques. Further, the symbolic representation of the effect can assume precise mathematical properties that permit greater precision in prediction. Thus, ambiguity in the discussion of the relationships between variables is reduced by precise measurement.

Statements like "high ambient temperatures may exert a detrimental influence on learning" carry global, but not precise, meaning. How "high" must the temperature be to exert this influence? Do variations in relative humidity combine with temperature differences to produce different effects? Does the clothing the person is wearing moderate these effects? (Or for that matter, determine what temperature is designated as "high"?) Is the length of time of exposure related to these effects? What is meant by the term "learning"? Does it mean something other than "performance"? Arousal levels are measured by self-report inventories as well as physiological arousal readings i.e., heart rate, blood pressure, or blood sugar level can be measured by some instruments. However, the problem here lies in the fact that the cause of these readings remains unknown. We do not know whether the rise in blood pressure is due to the environmental stimuli that we are studying or due to the personal health factors of an individual.

Second postulate:

2A: The environment, physical as well as social, typically exerts a steady-state influence on the behavior of its inhabitants. This assertion is not unlike the claim of Ittleson, Proshansky, Rivlin, and Winkel (1974) that the environment frequently operates below the level of awareness. The environment is seen as being taken for granted—we operate within an environmental context without paying much attention to it.

"Normative" or "normal" behavior occurs in these circumstances. We behave according to the cultural purpose of the setting. However, if the environment changes (or we change environments), we become aware of it because it is at that point that we must consciously begin to adapt or adjust.

Roger Barker's (1968) notion of "behavior settings" is also congruent with this assertion. The arrangement of chairs at a table (Sommer, 1974) or desks in a classroom (Sanders, 1958), for example, have very powerful but non-conscious effects on their users; in a very real sense, we inherit the use of space. Thus, the environment influences our behavior even when its physical and social characteristics are within a normal range, but under these conditions, we are typically unaware of that influence. Thus, you were probably not aware of the fact that on your first day in the room in which this class meets, your behavior of note-taking vs. ballroom dancing was determined, in part, by the setting itself. Nor did you probably give much thought to the fact that you were dancing instead of taking notes the last time you attended a nightclub.

2B. When the measured values of the environment's most salient characteristics undergo dramatic change, the environmental influence can no longer be characterized as a steady state. Helson's (1964) adaptation-level theory represents a general framework for the study of diverse responses to any set of stimuli varying along some dimension. Put briefly, Helson maintains that for any specified dimension of stimulus variation, individuals establish an adaptation level (AL; a preferred or expected level of stimulation), which determines their judgmental or evaluative response to a given stimulus located on that dimension. Deviations from AL in either direction (i.e., either increases or decreases in stimulation level) are evaluated positively within a certain range, but beyond these boundaries' changes are experienced as "unpleasant." There are also individual differences in levels of stimulation. For a person living in Jacobabad, a temperature of 35C might be normal because his body has adapted to the temperature of Jacobabad, and for a person living in Murree, a temperature of even 30C might be very high because his body has adapted to the temperature of Murree.

When environmental-stimulus properties change to the extent that they have exceeded the AL boundaries, the individual experiences "unpleasantness," and the influence of the environment is no longer steady-state. It is only at this point that changes in the normative modes of behavior can be expected, that the regularized, routinized behavioral-setting influence begins to break down.

If, as you sit reading these pages, your phone rings instead of the one four doors down the hall, or someone walks into the room wearing an unusual perfume (or no deodorant), or the room temperature increases to 85 degrees Fahrenheit or is reduced to 60 degrees Fahrenheit, or there is a knock on your door, then the environment no longer can be characterized as steady-state. Your response to the environment is now conscious and deliberate. You get up to answer the phone or the door; you look up to see who is wearing the unusual perfume; you open a window or turn up the thermostat (or put on or take off clothes) to return to a condition of thermal comfort. In other words, you become aware of your environment and begin adapting to it or making adjustments

to it to achieve a new equilibrium.

2C. Disruptions from steady-state occur when present perceptions do not correspond to desired or expected levels of physical and social stimulation.

مستحکم حالت سے رکاوٹیں اس وقت ہوتی ہیں جب موجودہ تاثرات جسمانی اور سماجی محرک کی مطلوبہ یا متوقع سطح کے مطابق نہیں۔ Disruptions in steady-state influence are brought about by the processes of sensation, perception, and cognition. Sensation involves those processes by which the world can be known to the perceiver. These processes typically involve the following modes: touch, smell, taste, vision, and hearing. We know silk by touching it, rotten eggs by smelling it, an orange by tasting it, and so forth. Another example would be two students. For a careless student, 5 hours a day of study can be too much and he might not be able to bear it but for a careful student hour, a day can be unto his satisfaction level. In other words, a careless student needs less study stimulation and a careful student needs more study stimulation.

Environmental cognition involves further processing of information (e.g., storing, organizing, and recalling). It also involves appraisal processes. Is this environment good or bad, cold or hot, strong or weak? It includes the emotional impact of environments, attitudes toward environments, the preferences we have for some environments over others, and the categories we use to organize our knowledge about various settings. Only human beings are capable of adding this element of likeness and dis-likeness, computers cannot do that. Through these various processes, we appraise environments and compare them with mental images of what we desire or expect them to be. Most of the time, because we have planned well, environments are acceptable approximations of what we expect. Under these conditions, the environment exerts a steady-state influence. Disruptions occur when we learn through these processes that present conditions do not correspond to our desired or expected levels of stimulation or are incapable of meeting the objectives of the plan with which we entered the situation.

We will start our next lecture by discussing the third postulate in detail

Lesson 8

The present framework of Environmental Psychology (II)

We will begin this lecture by discussing the third postulate of Environmental Psychology in detail.

Third postulate: The influence exerted by the environment is indirect i.e. Environment acts to influence people's emotional states, which in turn mediate their overt behavior.

ماحولیات کا اثر بالواسطہ ہے یعنی ماحول لوگوں کی جذباتی حالتوں پر اثر انداز ہونے کے لیے کام کرتا ہے، جس کے نتیجے میں ان کے ظاہری رویے میں ثالثی ہوتی ہے۔ The work by Baron and Bell also postulates emotion (affect) as a mediating link between the physical environment and aggressive behavior (e.g., Baron & Bell, 1976; Bell & Baron, 1976). It has been shown that such diverse environmental conditions as **temperature** (Griffitt, 1970; Griffitt & Veitch, 1971), **crowding** (Baum & Valins, 1973; Griffitt & Veitch, 1971; Valins & Baum, 1973), **noise** (Bull, et al., 1972; Geen & O'Neil, 1969; Mathews & Cannon, 1975), **air pollution** (Rotton, Barry,

Frey & Soler, 1976), and radio news broadcasts (Veitch, DeWood, & Bosko, 1977; Veitch & Griffitt, 1976) influence the affective state and interpersonal behaviors of individuals. If you feel annoyed by the phone ringing you might answer it in gruff tones, or not at all; if you are gladdened by the distraction, you will probably answer with greater civility. If you like the smell of an unusual perfume, you are likely to strike up a pleasant conversation; if instead what you are confronted with is the smell of stale cigarettes and body odor, your disposition is likely to be less shining, and your overt behaviors less positive.

Fourth postulate: Environment-evoked emotions are best depicted in terms of three distinct dimensions: pleasure – displeasure, degree of arousal, dominance –submissiveness. Russell and Mehrabian (1977) have provided evidence to reconcile these two sets of research findings and have reported that the three dimensions of pleasure-displeasure, degree of arousal, and dominance-submissiveness constitute both the necessary and sufficient dimensions to describe all emotions. (i) **Pleasure and Displeasure:**

For example, you remember a very sad event in your life. While thinking about that event you might experience pleasure or displeasure based on whether the event was happy i.e. winning a lottery or whether it was sad i.e. losing a mobile phone.

(ii) **Degree of arousal:**

When you remember the event in which you won the lottery, you might experience the same physical symptoms that you experienced when you won the lottery i.e., the same adrenaline rush, rise in blood pressure, or pupil dilation. جب آپ کو وہ واقعہ یاد آتا ہے جس میں آپ نے لٹری جیتنے کے وقت محسوس کی تھیں، تو آپ کو وہی جسمانی علامات کا سامنا کرنا پڑ سکتا ہے جو آپ نے لٹری جیتنے کے وقت محسوس کی تھیں،

(iii) **Dominance (control hona) -Submissiveness: (control na hona)**

Suppose you are a student who has topped the exam and after 6 months you remember this event. You will suddenly feel that in that particular situation, you might have achieved anything (you will generalize your self-confidence to other situations), and you might start to realize how much control you were able to exert over the environment or how much power you were able to exercise. آپ کو یہ احساس ہونا شروع ہو سکتا ہے کہ آپ ماحول پر کتنا کنٹرول رکھنے کے قابل تھے یا آپ کتنی طاقت استعمال کرنے کے قابل تھے۔ When you attach this degree of control to the high level of arousal and attach this whole experience with the element of pleasantness, you will feel excited, as if you are dominating the environment. Now imagine the opposite, where you are a student who has failed, you will simply think that everything that happened, is because of external environmental factors. You will feel that you were not able to control the environment and ultimately feel your submissiveness at the hands of the environment.

Results of their studies show that the larger number of dimensions obtained in verbal-report studies can be accounted for by these three dimensions. Thus, self-reports of emotions that employ different words (e.g., joy and happiness) may yield similar underlying configurations of pleasure, arousal, and dominance, which differ primarily in the intensity of affect. Further, differences between global emotional states can be best understood in terms of differences in the underlying configuration of these three dimensions. Thus, the best available evidence to date suggests that the emotional (affective) state of individuals can be adequately described by these three dimensions.

Fifth postulate: The environment-influenced behavior of the individual is dependent upon the extent and the configuration of the dimensions of emotions aroused. In most of the studies to date where the emotional state of the individual is seen as a mediating factor for behavior, emotion has been conceptualized in a unidimensional manner. Typically, the one dimension utilized has been pleasure-displeasure, although sometimes a variant of the dominance/submissiveness dimension has been used (e.g., studies of control, constraint, and reactance). However, researchers have seldom considered the interrelationships of these dimensions. The current framework attempts to utilize a multiple-dimension approach to emotion as a way of accounting for more of the reliable variance in the relationship between emotional states and overt behavior.

We would expect different behaviors to ensue as a result of changes in the environment that elicited displeasure, high arousal, and submissiveness ("anxiety") changes resulting in displeasure, high arousal, and dominance ("anger"). A teacher experiencing the former, for example, may attempt to "leave the field," whereas experiencing the latter may lead him or her to combat, or at the very least, to attempt to change the environment. Aggression and excitement have the same physiological arousal but their experience is categorized differently. Re-living the experience is also different from the actual experience.

Sixth postulate: Behavior is goal-directed. That is to say behavior in an environmental setting is performed to increase or decrease the pleasure, degree of arousal, or dominance elicited by changes in the environment in an attempt to return the environmental influence to a steady state. Behavior is a way to get your goals. underlying assumption is that steady-state environments are preferred and actively sought (i.e., humans don't passively react to environmental stimulation, but rather attempt to alter their experiences to maintain a state of equilibrium). They do this by altering their evaluation, by altering the setting, or by changing settings altogether. For example, a student who completes a very difficult assignment by considering work from an angle of entertainment.

In response to displeasure aroused by stimuli whose preferred values are exceeded (in either direction), the behaving organism attempts to decrease the unpleasantness. For example, the temperature of a particular environmental setting exceeding AL (adaptation level) elicits discomfort. Simultaneously, the degree of arousal and dominance/submissiveness will also be influenced. If the organism experiences great displeasure, high arousal, and high dominance, then aggressive behaviors may occur, which might be directed at removing the social source of that emotion. On the other hand, if displeasure is not felt, if the degree of arousal is low, or if submissiveness is felt, submissive behavior might occur. This could involve withdrawal to a different setting more conducive to the desired emotional configuration.

Seventh postulate: When goal-directed behavior is ineffective or when great expenditures of energy are required to maintain a steady state, the environment can be considered pathological and a disruptive influence on human functioning. جب اہداف پر مبنی رویہ غیر موثر ہوتا ہے یا جب ایک مستحکم حالت کو برقرار رکھنے کے لیے توانائی کے بڑے اخراجات کی

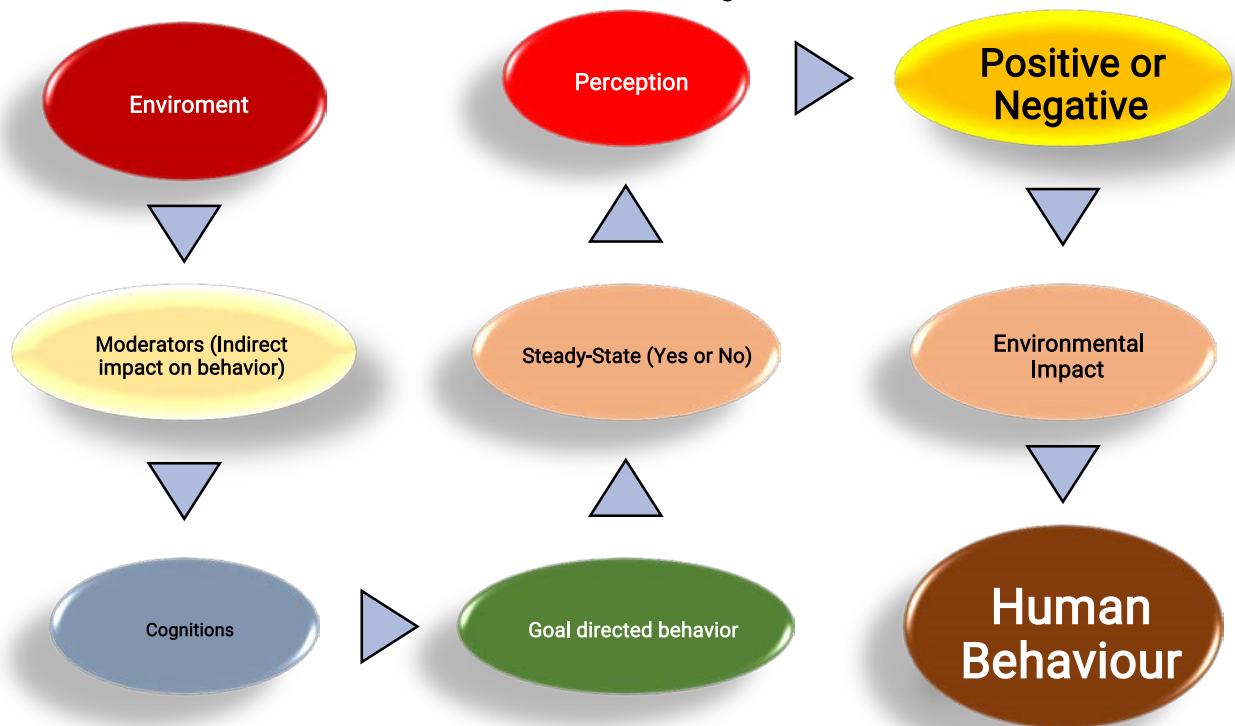
ضرورت ہوتی ہے، تو ماحول کو پیٹھولوجیکل اور انسانی کام کاج پر ایک خلل ڈالنے والا اثر سمجھا جا سکتا ہے۔ جب مسلسل کوششوں کے بعد بھی انسان اپنا گول حاصل نہیں کر پاتا تو وہ ڈیپریشن میں چلا جاتا ہے۔

What are the long-range effects of exposure to a given environment featured by a particular level of intensity, complexity, and inappropriateness of stimulation? According to adaptation level theory, the individual's AL will be shifted to a value corresponding more nearly to that of the environment. This, of course, is what adaptation is. However, for adaptable humans, it is still possible that the range of environmental stimulation could be such that the energies expended in adaptation would have detrimental effects on the individual, or that the behavioral repertoire of the individual is too limited, thereby making adaptation impossible. It is under these conditions that the environment can be considered pathological. Evidence (to be detailed in later chapters) concerning the effects of prolonged noise exposure (e.g., Glass & Singer, 1972; Weybrew, 1967) exemplify this state of affairs, as does the work of Calhoun (1962) who looked at the long-term effects of increased population density on social adaptive behaviors of Norway rats. Another example can be year-long desert or winter survival courses. The excitement for adventurers is usually high on such training but with time the excitement level dies down and survival becomes difficult.

Model: How do the seven postulates of the environmental framework interact with each other?

Environment →	Moderators →	Cognition →	Steady State
- Noise - Pollution - Weather - Crowing - Mountains/Deserts - Architecture	Social situations: - Relationships between different people. - Classroom settings. Personal Variables: - Adaptation level. (A person in Murree will feel a temperature of 35°C differently than a person in Jacobabad due to their different adaptation levels of temperature)	- Expectations. - Attaching emotional meanings to environmental stimulus. - Goals - Moods - Sentiments - Personal expressions	What is the effect of the environment on the human? Is it helping them gain a steady state or is causing disruptions in their behavior?

Let's understand this table in the form of a reactive diagram:



This course has also been divided into chapters. Chapter one consists of (Lesson1-Lesson8).

From the next lecture we will start chapter2.

Lesson 9

Chapter 2: Environmental Perception and the use of natural environment.

Lesson 9: Elementary Psychophysics/Perception and its Cognitive Bases

In elementary psychophysics, we try to understand how human beings can psychologically detect physical stimuli (Stimulus Detection). For example, if a student is deeply involved in studies, he might not notice any other sound in the environment such as the sound of the fan in his room or the ticking of the clock. Elementary psychophysics tries to understand at what time and how physical stimuli gather such momentum, that they can be interpreted psychologically by a person. Perception is a basic tool which helps us to learn about our environment and to design our responses

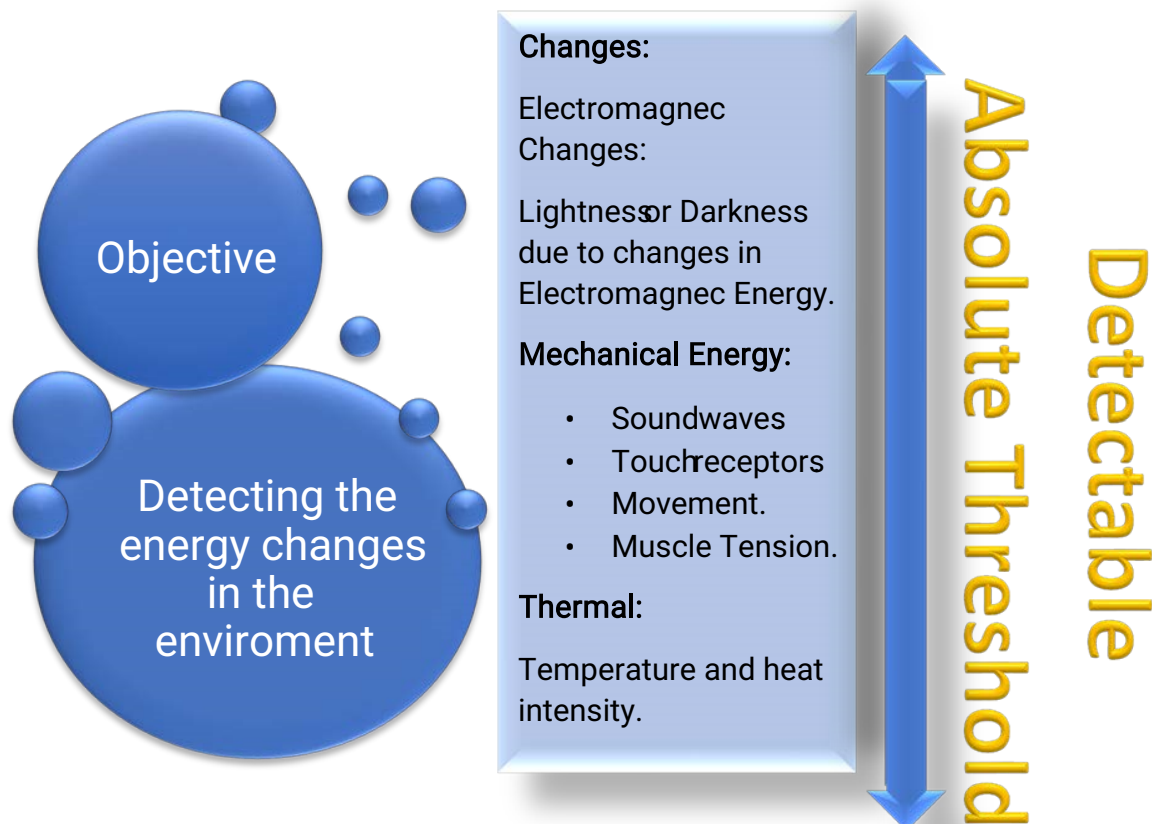
Objectives of Elementary Psychophysics:

There are four objectives of Elementary Psychophysics:

1. Stimulus Detection:

What is it? For example, suppose that you are walking through a dark street at night. At

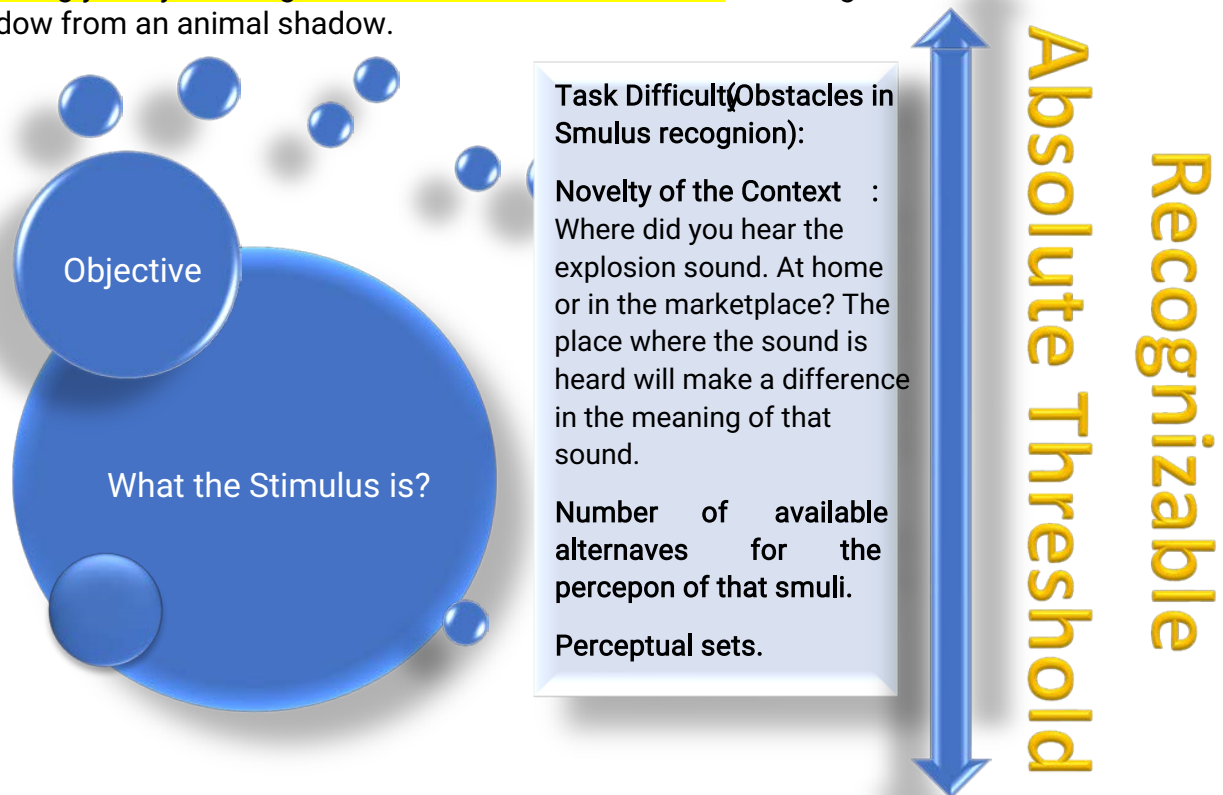
the end of the dark street, you saw something that felt like the figure of an animal. But you are not too sure about it because light is very little. You might dilate your pupils or try to use your depth perception cues in an attempt to detect the stimuli. The basic objective of stimulus detection is to understand what is happening around us. In Psychophysics this relates to the different types of energy changes that are taking place in the environment. Our receptors can read those energy changes.



Every stimulus is detectable up to a certain point but beyond that range it becomes undetectable. All five senses have their ranges in between which they can be detected. The minimum intensity required by a stimulus to be detected is called the absolute threshold.

2. Stimulus Recognition:

Is this stimulation different from that one? For example, when you have accommodated your physiological systems and sensory receptors according to darkness then that information goes inside your brain. That is the time when you perceive those current images and **you match those images to the previous images of such things that exist in your mind and then as a result of that matching you try to recognize what the stimulus is about** i.e. being able to differentiate a human shadow from an animal shadow.



There are also minimum or maximum ranges for stimuli to become recognizable i.e., is this the picture of an animal or a tree. For example, you are studying and you hear a loud sound outside your home. Now you have to recognize whether it is the sound of a car crash, a cylinder blast or an actual blast. The novelty of the context and available meanings you can give to the explosion sound in that particular situation determines how you can recognize the explosion sound.

3. Stimulus discrimination:

How to differentiate between different stimuli? For example, you may need to differentiate between similar stimuli to come to the final decision of what that stimulus actually is. In the explosion sound example, **you may think of the sound of a car crash, cylinder blast or bomb blast and do a comparative analysis of these sounds in your mind to make the final decision about what sound it was.** This brings us to the concept of Just noticeable difference (JND):

Just noticeable difference (JND): An increase in the intensity of a stimulus above its original level required for a person to notice a difference in the level of intensity. The intensity of the sound will increase or decrease concerning the person's distance from the place where the sound is coming from. A person standing near the explosion sound will have more requisite information to perceive that sound accurately/to create such a noticeable difference as compared to the person who was far from the explosion sound.

Weber's Law:

High-intensity stimuli require a proportionately larger increase to produce the Just noticeable difference (JND). For example, if you are listening to music in less volume, even a little increase in the volume will be noticed by you but if you are already listening to loud music, a little increase in the volume will not be noticed by you. You need to increase the loud music much more to create the Just noticeable difference between the two loud music sounds.

4. Stimulus Scaling:

How much of a certain stimulus is there? For example, when you are passing through a dark street and you look at something that looks like the figure of an animal, scaling would be when you try to find out that is the animal big or small and what type of animal this is. Then you would appropriately design your reaction towards it. It is a process for effective commerce with the environment. The magnitude of the stimulus, as scaled by the perceiver, determines the magnitude of the response deemed necessary to deal with it. For example, if the sound of the explosion is too loud, the person standing near the explosion sound will have a different reaction according to it. The same is the case with low-intensity explosion sound.

Perception

Perception means to look at, to understand, and to know about the world around us. Perception is a sudden realization or an insight experience of our ability to navigate the environment on our own. For example, when a 5-year-old child rides a bicycle for the first time in life without the support of his/her father, he is suddenly surprised and starts to realize how much he/she is capable of. When we think of perception we usually think first of visual perception. This is probably because we gather so much information from the environment via this sensory system. Except listening to speech, the vast majority of our everyday activity is guided by vision. Vision guides our motor behavior (getting us from here to there while avoiding running into things on the way), providing us with information concerning what is out there and where it is. Additionally, much of our interpersonal behavior is guided by the visual information we receive. Our judgments of the emotional states of others (and ourselves, for that matter), and our intentions, likes, and dislikes, are determined in good part by what we see. No wonder we rely so heavily on our sense of vision to inform us of the world that we inhabit! There is an old adage that "seeing is believing," and when it comes to conflicting cross-modal information, we do indeed tend to believe our eyes over the other senses.

Despite this great reliance on vision, perception is more than just sensory input to the visual system. Often, we must depend on other systems when interacting with the environment. For example, we cannot see heat, so we must depend on thermal receptors on the skin's surface to warn of the dangers of placing the hand on a stove; we cannot see natural gas and must depend on olfactory cues and odorous additives to warn of a gas leak in the home; we cannot see the Civil Defense siren warning of an impending tornado, and therefore must rely on our sense of hearing. But **perception of the environment** is even more than the summation of all these sensory inputs. It **involves labeling, describing, and attaching meaning to the world** around us. Perception, in addition to being sensory, is also highly cognitive.

(i) **Labeling:**

It means to identify what is happening around us. Or what the stimulus is about? For example, after looking at a smiling facial expression, **we can identify is it a happy smile or a sarcastic طنزيه smile.** (ii) **Describing:**

It is the interpretation of details about the context of the stimuli. For example, if we know that a person is smiling because of happiness, our next step would be to look for further details i.e. **why is the person happy?** Is he happy because he received an increase in his salary or is he happy because he is going on a vacation?

(iii) **Attaching meaning to the world around us:**

It means to make the world around us meaningful. For example, while teaching a child how to cross a road, we also need to teach him the dangerous outcome i.e. fatal accidents that may happen if he does not follow the road rules. The whole experience gives a bulk of information to the child at one time. In other words, we need to go beyond then the present stimuli so that a person can make sense of perceptual experiences.

Environmental Perception (Cognitive Bases of Perception):

All environments carry a set of meanings acquired through their specific physical, social, cultural, aesthetic, and economic attributes. Let's look at them one by one:

An example of a beautiful picture of a waterfall with one-man swimming in it.

- (i) **Physical attributes:** This means the physical setting of the stimuli. For example, upon looking at a beautiful picture of a waterfall, **we might focus on the color of the water, the plants on the waterside, or the mountains behind it.**
- (ii) **Social Attributes:** This is the social scenario around the perception of the stimuli. For example, **if only one person is diving in the waterfall, a happy person might say that this person seems peaceful and happy enjoying himself and a lonely person might think what a lonely person he is! Diving alone in the waterfall.**
- (iii) **Cultural attributes:** This is the content of the stimuli in relevance to the person perceiving it. For example, **while looking at the waterfall picture we might question ourselves that does this waterfall exists in Pakistan or if it is a scene from a western country.**
- (iv) **Aesthetic attributes:** **These are the personal ways we find beauty and meaning in stimuli.** For example, a person with a high aesthetic sense might look at the picture of the waterfall **and think "I feel full of life when I look at this picture"** or **"I feel like revisiting my childhood after looking at this picture".**

- (v) **Economic Attributes:** This is the assessment of economic resources required to interact with the stimulus. For example, a man might look at the picture of the waterfall and think "How can I visit this waterfall", "How much expense will it take to dive into this waterfall" "How much travel cost will it take to visit this waterfall. For example, a Pakistani person might immediately think of travel costs if he looks at the picture of Niagara Falls which is located in Canada.

These meanings are extracted from the environment by the perceiver in terms of his or her attitudes, beliefs, values, and physical limitations. We may admire an apple orchard in spring for its floral beauty and its aromatic fragrance, while simultaneously realizing its worth in terms of the honey that will be produced by bees from its nectar and the apples to be harvested in the fall. Additionally, we may see the orchard as symbolic of the economic power of its owner, his or her ability to buy and sell, and ultimately the political influence that person is likely to exert in the community. Finally, we may see it as representing the outcome of years of research in the development of hybrid apples. We perceive all of these meanings and respond to them to some degree as the sight and smell of the spring blossoms reach our eyes and noses.

► **Assessment of Environment:**

Our assessment of the environment is achieved within the context of three broad but not always congruent ways of viewing the world.

- (i) **Culture:** First, we develop attitudes as a result of living within a culture. For example, in Pakistan, the culture is different from one city to another city. Our culture defines our norms and values, which, in turn, dictate our general behavioral patterns. As we studied previously nomadic cultures promote independence and how agricultural culture promotes obedience. Now, a person from a nomadic culture, when visiting an agricultural societal environment, might assess the environment as being "suffocating" or "controlling" for its rules like obedience and staying in one place for a long time.
- (ii) **Needs and Preferences:** We also assess the environment according to our needs and preferences and behave accordingly. For example, imagine that you are feeling very hungry and the food is not ready. Now, this frustration caused by hunger and nonavailability of food will provoke you to think "What a bad day it is today" and you might start watching cooking shows on TV while waiting for your food. Cooking shows also come on TV at specific times before or near dinner time to get more views.
- (iii) **Future:** We also assess our environment in terms of how it will affect us in the future. For example, a housewife might think of separating parents' room from their children's room because near future children will grow up and require their personal space. Not only is the question of "What's in it for me?" asked, but also, "What effect will my presence and interaction have on the environment being viewed?" If trees are seen only in terms of the apples they will produce and care is not taken to preserve them, soon there will be no apples and eventually no trees. This view of the environment is much like the ecologist's view.

Contextual and Social Bases of Perception (perception is different in different cultures ,also different according gender and society .)

It is important to emphasize that perception is contextual. Cultural, social, gender, and individual differences all influence what we do and do not see in our environment. Looking within the context of gender, there are characteristic differences in the ways in which males and females tend to appreciate certain things. This is the reason that cars are designed differently for males and females. For instance, consider how exposure to green spaces (natural environment) versus urban concrete jungles (built environment) affects people's stress levels, attention, and overall well-being. Another example would be how an individual's identification with larger social units (such as community, nation, or world) strengthens in-group solidarity and empathy, influencing their support for environmental protection.

Thus, perception is not simply a matter of the individual responding to sensations created by energy from stimuli impinging on the sensory organs. Rather, this process is embedded in a cultural context, and various social factors have been demonstrated to produce differences in the ways two individuals perceive the same stimulus. Individual differences in backgrounds, experiences, values, and purposes can have a profound influence on the result of the processing of information from the world around us. These differences, however, do not detract from the fact that perception is a fundamental psychological process in which all humans engage.

Lesson 10

Perception and its Cognitive Bases/Theories of Environmental Perception

Complex Perceptual Processes

The exercise below illustrates the complexity of processes taken for granted and engaged almost automatically when perceiving individual stimuli in the environment. As indicated, such processes are critical, and psychologists have exerted considerable effort to understand them. However, the primary concern of environmental psychologists is understanding how people perceive molar aspects of the environment. In addition to understanding how discrete stimuli or objects are perceived, environmental psychologists attempt to understand the processes by which whole environments are perceived. In a sense, they are interested in the forest, not just the individual trees. Further, a discussion of these processes will follow since it is important to account for the relationships among the perceptual, cognitive, and evaluative processes that affect experience and determine behavior. Although highly interdependent, they will be discussed separately, making note of their interrelationships where appropriate.

► Exercise:

Below is a saying by Masaru Ibuka, a Japanese man who was the founder of an Electronic Company, in 1991.

We worked furiously سے غصے [to realize our goals]. Because we didn't fear, we could do

something drastic.

When you initially read this saying it almost seems like an automatic activity. But from a research point of view, the human might go through the following **complex perceptual processes** when reading the above saying.

1. We Start reading it with a specific purpose.
2. We Screen out other stimuli.
3. Our Eyes translate the light waves reflected off to screen into neural energy.
4. Our sensory pathways transmit a series of electrical impulses to the higher brain centers.
5. The brain organizes and interprets the input.
6. The brain matches the present patterns of sensation with previous images.
7. Finally, Words and phrases became coherent and meaningful.

Theories of Environmental Perception

► Gestalt Theory:

Much research and theory exist regarding the basic processes involved in perception. Among the earliest and most significant contributors to this area was a group of German psychologists working within a framework known as Gestalt theory (e.g., Koffka, 1935; Kohler, 1929). **These theorists emphasized the active role of the brain in searching for meaning in stimuli.** (i) **The Principle of Good Form:**

The term Gestalt means "good form" and Gestalt theorists proposed that the brain is organized in such a way as to construct meaning from stimuli, and even to impose meaning where it might not appear to exist objectively to achieve this "good form. For example, read the following sentence:

THE QUAK BRON FAX JUNS OUER THA LEZY DAG

Many people can read this sentence correctly as "*The quick brown fox jumps over the lazy dog*". It is because of the principle of good form that our brain constructs meanings from any stimuli even where they might not appear to exist. This entire sentence was incorrect in spelling but we were still able to read it correctly. Good Form has its benefits as when it helps us to make sense of even a disorganized environment, however, it is negative in a few places as well. For example, a teacher might mark wrong answers as correct due to good form and students might receive unjustified benefits. Good form may be harmful or even dangerous if it interferes while performing detail-orientated tasks like performing heart surgery. Any little mistake to wrong meaning construction by our brain due to good form can lead to heart failure. Due to this good form may also be called a "False Positive" or False Negative". It is not uncommon for students, when proofreading their term papers, to "read" words that do not exist on the paper. The sentence "The bear climbed up *a* tree" might be read, "The bear climbed up *the* tree."

(ii) The Holistic Orientation:

A second characteristic of the Gestalt approach is its holistic orientation (i.e., **the assumption that the perceptual process must be understood in its entirety rather than broken down into discrete elements**). This assumption is often expressed in the statement that in perception, "the whole is greater than the sum of the parts." For example, we do not say "I am watching a show on a box which has a screen, buttons,

and a black outline” We simply say “I am watching a show on TV”. We perceive things as a whole in one part. Look at the following figure:



We know that this image is that of a star (because of good form) even though there are some blank spaces left in it. Good form does not only imply physical stimuli but applies to our cognitive processes as well. For example, a student who has passed college remembers her school days as the best days of her life. However, she might not remember every detail. She forms a complete image of her school time as a happy image, ignoring the little positives or negatives that come along the way.

(iii) Size Constancy:

Another illustration of the tendency to impose meaning on sensory inputs is the principle of size constancy, in which changes in the size of the retinal image of an object are interpreted as changes in distance rather than changes in the size of the object. In this case, the brain is actually "overriding" sensory inputs to maintain a Gestalt. For example, look at the following image:



Now, due to the principle of size constancy, you might perceive that the hand on the left side is far away and the hand on the right side is near to us. In a literal sense, the size of both hands is different. However, our brain knows that they are the hands of the same person so they will be the same.

► Functionalism:

A theoretical orientation that differs from the previous approach views perception as a much more direct process. It involves less mediation by higher brain centers to perceive meaning in the environment.

1. It is argued that meaning already exists in the environment and that our sensory mechanisms are "prewired" to respond to meaningful aspects of our environment. For example, due to our past experiences, we might assume the content of the book just by looking at its title. Another example is the generational experience example. Due to the experiences of our ancestors, we naturally perceive our environment. We have pre-formed meaning. So, there is no need to use higher brain centers and perception occurs in sensory areas or organs directly. Due to this, the perception process seems automatic.
2. This approach is related to **ecological biology**, which studies organisms' adaptation to their environment. For example, the concept of an **ecological niche** refers to the instinctive tendency observed in animals to seek out that area of their environment that affords them the greatest chance of survival. For example, animals have the same methods of hunting that they used 100 years ago. Due to this reason, tigers live in environments where there is a widespread landscape and other animals are co-habituating so that they have plenty of area to search and hunt and there is plenty of prey, ultimately, increasing their chances of healthy survival. But this is not true for humans as our behavior constantly keeps on changing. For example, the working ratio in today's world is different than it was in the time of our grandparents. They used to work in the day and the rest of their time was for their families. However, nowadays people do not have much time for their families, and multitasking is highly appreciated. There was a time when a movement named **"Ethology"** started in Psychology. In this movement, **psychologists attempted to study instinctual behavior in humans**. For example, the instinct of aggression or the will to power. But with time it became evident that humans cannot be studied in terms of instincts. Because we know that instinctual behavior is constant, and human behavior keeps on changing.
3. Gibson (1979) applied this notion to human perception. She suggests that **humans are innately endowed with the ability to perceive those aspects of our environment that have functional value for them**. Thus, according to this view, an infant should be born with the ability to perceive its mother's face, since this stimulus has obvious survival value. Because as compared to animal babies, human babies depend a lot on their mothers to survive and to learn to function in their environment. In other words, mothers have a high functional value for children and most children recognize their mothers as soon as they are born.

Lesson 11

Learning Theory/Probabilistic Functionalism and Environmental Cognition

Learning Theory:

Much research on human perception concerns the role of experience in perceptual development. From this perspective, our perceptions are not innately determined, but rather we must learn to perceive critical aspects of our environment. For example, the principle of size constancy referred to above is not seen as an inborn perceptual ability, but one that develops only through the experience of seeing many objects from a variety of distances. Gradually the infant learns that the objects are not growing or shrinking, but remaining a constant size regardless of their distance. This happens even though the retinal images, and thus the neural impulses sent to the brain, vary dramatically.

Gestalt and Learning perspectives have a lot in common. According to the Gestalt theory, Perception is done almost within seconds along with the involvement of higher brain centers because it holds the belief that perception is done directly via sensory organs and receptors as well as with the help of information we hold from past experiences. Learning theory also places a lot of emphasis on direct information processing, past experiences, and specialized interpretation of information. Functionalists on the other hand are different in the sense that they do not focus on the involvement of higher brain centers and place more emphasis on mechanical instinctual behavior.

Learning theorists propose that an important result of experience and learning in perception is the development of assumptions about the world around us. For example, youngsters might look at cool and handsome heroes in movies. They might develop the assumption that cigarette smoking will make us look like a man, it will make us look stylish. Due to this assumption, they might start behaving similarly and start to smoke. However, this assumption is quite misleading, and usually such youngsters end up facing the hazardous effects of smoking on health.

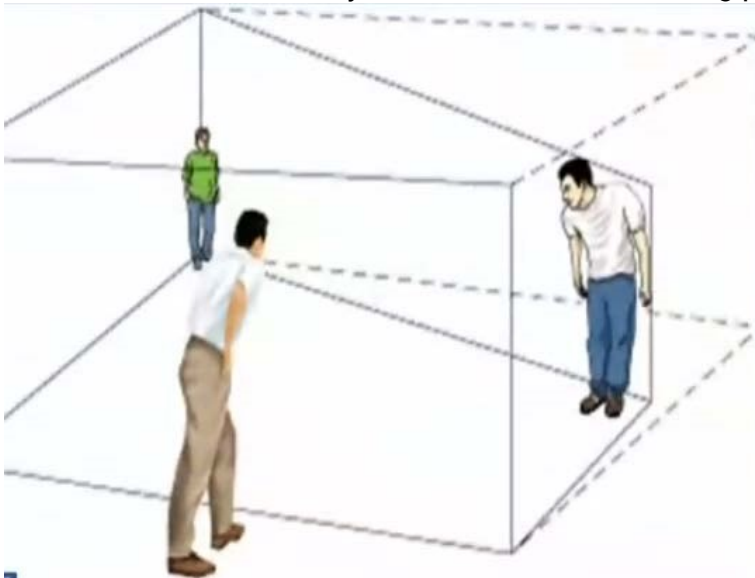
These assumptions facilitate our interactions with the environment because they save us time and effort in coping with new stimuli. That is, we do not have to approach new situations as though we have never encountered them before. We assume that many elements of the situation are similar to those of situations that we have previously experienced. For example, our responses to the word mother are usually like "Mothers are kind" or "I love my mother". In other words, they are our assumptions about the word mother. Gestalt Psychologists also embrace the concept of assumptions because according to them perception process is very fast. Sensory receptors might process information in seconds and they believe that it is assumptions that make perception process faster.

Thus, we bring to the present situation learned assumptions in the form of expectancies about what is likely to happen. These expectancies are usually correct, making for easy processing of information and adaptive functioning. In these ways, learning theories are not unlike those of the functionalists.

Our assumptions, however, can sometimes be misleading, resulting in misperceptions or illusions (e.g., Ames, 1951). For example, look at the following image:



By looking at this picture you might think that the person on the right-hand side is bigger and the person on the left-hand side is a kid. However, this is just an optical illusion and has no contact with reality. Now, look at the following picture:



This is the actual reality of the people shown in the first picture. The room is designed in such a way that the backspace is a bit tilted. It naturally makes the person in the white shirt stand a bit forward and bent down so he appears bigger. And it also naturally makes the other person in the green shirt stand at the back, making him look smaller like a kid.

Probabilistic Functionalism

Given the complexity of environments, the goal stated earlier that environmental psychologists must account for the processes by which molar environments are perceived is an exceedingly difficult one to achieve. Therefore, one of the most important learning theories of perception, and potentially the most fruitful for environmental psychology, is Brunswick's lens model (1956, 1969). Brunswick's approach provides a model for mathematically describing individuals' perceptual processes when making judgments in response to molar environments containing multiple stimulus dimensions. Let's look at the following figure:

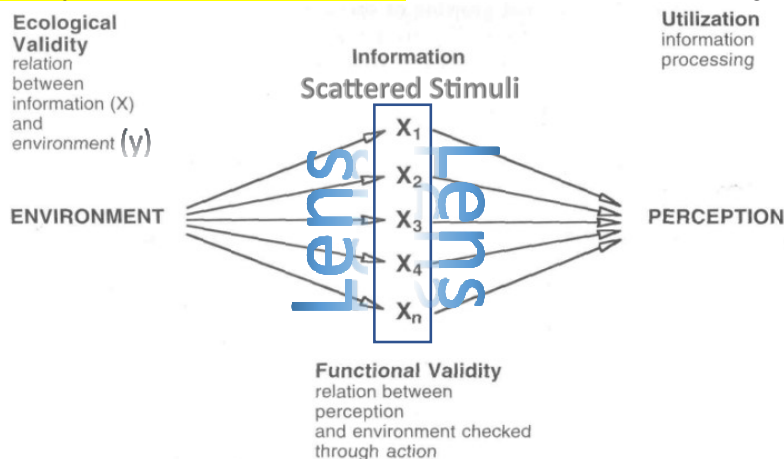


FIGURE: Brunswick's probabilistic theory illustrates one way of relating the information available from the environment to the way the individual perceives the environment. Let's break down a few concepts of this diagram.

- **Ecological validity:** It is what information we focus on. It shows the relationship between information x and environment y . It means in a certain environment; a certain stimulus is more important and our perceptual lens would automatically concentrate upon that particular information.
- **Functional validity:** The information on which our perceptual lens focuses has certain functional validity. It means the relation between perception and environment that is checked through action. For example, children learn to walk in a certain style or by using certain balance mechanisms. Children's' perceptual lens focuses on these techniques which have helped them to learn to walk.

Brunswick argues that complex stimulus patterns are processed as though through a "lens" (see Figure above), where the scattered stimuli are "focused" into a single perception of the environment. An example would be being stuck in traffic where the environment is giving a bulk of information and our perceptual lens is focused on the information that is important to us. In this way we manage to reduce the complexity of our environment by filtering the various available stimuli through the lens, discounting or ignoring some perceptual cues while emphasizing others. We do this not only to simplify judgmental tasks, but because we learn from experience that some sensory information is misleading (in Brunswick's terms, they are lacking in ecological validity), and other cues are of minimal value, (i.e., of low ecological validity) in correctly

perceiving the true state of the environment.

Environmental Cognition:

All previous theories were about environmental perception. However, now we are going to discuss Environmental Cognition. Let's first look at the difference between the two:

► Difference between Perception and Cognition:

Perception is the process in which we use our sense organs to gather information about the environment and that information gets into our mind and helps us direct our behavior. For example, imagine you're walking in a garden and you see a rose. You notice its deep red color and its sweet fragrance. This immediate sensory experience of the rose is an example of perception. **Cognition** is a step ahead of perception which means that we are making a proper sense of the environment over here. The packets of incoming information here become a part of our working strategy, helping us build our insights. For example, Now, suppose you start thinking about the rose, recalling memories of roses you've given or received in the past, or considering the symbolic meanings of roses in different cultures. This process of thinking, understanding, and interpreting the rose is an example of cognition.

The term, cognition, refers to thought processes. Thus, environmental cognition concerns thinking about the environment (i.e., how individuals process information and organize their knowledge about the characteristics of their environment). Also, of interest is how this knowledge is acquired, or learned, as well as how individuals differ in their knowledge of the environment. Finally, environmental cognition concerns how variations in the environment affect the ability to understand the setting. For example, important differences in knowledge exist between familiar versus novel environments. We will consider novel environments first.

► Responses to Novel Environments:

From the participant's point of view, the environment is typically neutral and enters into awareness only when it deviates from some adaptation level. Your first few days on campus can be thought of as a non-neutral situation, one that from your perspective at the time deviated from the norm and made you self-consciously aware of your new surroundings. Ittleson, Proshansky, Rivlin, & Winkel (1974) claim that in situations of this kind, **six different but interrelated types of responses occur: affect, orientation, categorization, systemization, manipulation, and encoding.** While in the following paragraphs, we will discuss each of these in the order just presented, these responses are probably not made serially. Each tends to blend with the other as we attempt to bring the environment and ourselves into a state of equilibrium.

- ❖ **Affect:** (Nai jgah ka asar „arousal high ho jata ha)The nature of our affective response to new environments will depend on many factors, some of which will be discussed later in this chapter. At a minimum, the effect will consist of a heightened degree of awareness and arousal occasioned by the need to know, predict, and therefore to feel in control of and secure in an unknown setting. **Affect is our emotional response.** For example, you might enter into a new

building and say that I feel happy/secure here or I feel sad/insecure here. Aside from this general reaction, several other feelings may emerge because of the particular characteristics of a new setting. Discovering that the tennis courts are close to the dorm may lead to happiness, whereas finding that you live adjacent to the cemetery might lead to fear and nervousness. The thought of long walks to the library through a grove of oak trees might conjure romantic images and feelings of love, or graven images of rapists and muggers, leading to fear and anxiety. Such affective responses, both general and specific, may govern the direction that subsequent relations with the environment will take. First impressions (feelings) about places can and do have enduring consequences.

- ❖ **Orientation.** Individuals in a new setting actively seek to find their place, their niche. This is primarily a cognitive process and involves scoping out a place. For example, a student goes to the USA for studies. On his first day in university, the excitement will be followed by insecurity which prompts اشارہ کرتا ہے him to learn about his environment so that he can find out his role in it and adjust to it.

Six steps come under our responses to novel environments: *affect, orientation, categorization, systemization, manipulation, and encoding*. Affect and Orientation have been discussed in this lecture however all six processes will also be discussed in the next lecture again in detail.

Lesson # 12

RESPONSES TO NOVEL

ENVIRONMENTS AND ENVIRONMENTAL COGNITION

Responses to Novel Environments

A novel environment is an unfamiliar or new setting that a person encounters for the first time or after a long period of absence. Ittleson, Proshansky, Rivlin, & Winkel (1974) claim that in situations of this kind, six different but interrelated types of responses occur: affect, orientation, categorization, systemization, manipulation, encoding. While in the following paragraphs we will discuss each of these in the order just presented, these responses are probably not made serially. In fact, each tends to blend with the other as we attempt to bring the environment and ourselves into a state of equilibrium.

Affect

The nature of our affective response to new environments will depend on many factors, some of which will be discussed later in this chapter. At a minimum, the affect will consist of a heightened degree of awareness and arousal induced by the need to know, predict, and therefore to feel in control of and secure in an unknown setting. Aside from this general reaction, a number of other feelings may emerge because of the particular characteristics of a new setting. Discovering that the tennis courts are close to the dorm may lead to happiness, whereas finding that you live adjacent to the cemetery might lead to fear and nervousness. Such affective responses, both general and specific, may govern the direction that subsequent relations with the environment will take. First impressions (feelings) about places can and do have enduring consequences.

Orientation

Individuals in a new setting actively seek to find their place. This is primarily a cognitive process and to use a slang expression involves "scoping out" a place. You probably asked yourself on those first days: Where, relative to my dorm, is the dining hall, the recreation center, the library? Where do you go to buy books, to get athletic tickets, to pay parking fees? Which dorm is the one that your cousin Fred lives in? Ittleson, et al. (1974) describe the process of finding answers to these questions. Orientation, in short, expresses a person's desire to "know where one is" physically in relation to the total milieu. Here we see the beginning of environmental cognition, with the individual actively attempting to identify the location of important stimuli in the new environment in relation to where one is at.

Categorization

In new situations, though, a person does more than just orient; he or she also *categorizes*. The individual evaluates the new environment and imposes his or her own unique meaning to various aspects of it. Not only do people ask where they can get a pizza, but they also ask where to find the best or the cheapest pizza. Categorization is therefore the process of extending the meaning of the environment by functionally relating its various aspects to one's own needs, predispositions, and values. Thus, categorization represents a more sophisticated understanding of the

environment than simple orientation, in that the individual now knows several instances of stimuli in a particular category and is able to distinguish among them in terms of their relative utility to the satisfaction of one's needs.

Systemization

It is difficult to say where categorization leaves off and systemization begins, but at some point individuals organize their environments into more meaningful and more complex structures. They know, for example, the best time to go to the library, not only in terms of when it is the quietest, but also when it is easiest to find a place to park, or when the most helpful librarians are working, or when the latest issue of *Sports Illustrated* arrives.

Manipulation

Out of such systemization, individuals achieve a sense of order and understanding; they not only know the new setting but they can predict it and make it work to their benefit. If people have ordered their environment, they usually can manipulate it or control it to their advantage. For example, If the cheapest pizza parlor in town is closed, they know how to get to the second cheapest, or know their options with respect to having Chinese or Mexican food instead.

Encoding

Encoding represents the highest level of understanding about the environment, because the individual is no longer tied to concrete perceptions of the setting. People can function more effectively with this knowledge, and can also communicate with others about using shared symbols. Encoding allows us to "think-travel" through environments (see Figure 2-5) and to prepare us for changes in our interrelationships with our environments.

Characteristics of Cognitive Maps

A cognitive map is a mental representation of the physical environment. It involves perceiving, storing, processing, and recalling spatial information using our brain's cognitive abilities. It is a vital skill that helps us navigate and interact with our surroundings. A common approach to studying spatial cognition is to ask people to draw "sketch maps" of environments. Research indicates that sketch maps are a reliable method of data collection (Blades, 1990). Lynch (1960) conducted one of the first comprehensive studies of the nature of cognitive maps when he asked residents of three American cities (Boston, Los Angeles, and Jersey City) to draw maps of their city environments. He analyzed these drawings for commonalities in the features of people's mental images of their cities. This resulted in the identification of five major characteristics:

- 1- Paths
- 2- Edges

- 3- Districts
- 4- Nodes
- 5- Landmarks

Lesson # 13

CHARACTERISTICS OF COGNITIVE MAPS

A common approach to studying spatial cognition is to ask people to draw "sketch maps" of environments. Research indicates that sketch maps are a reliable method of data collection (Blades, 1990). Lynch (1960) conducted one of the first comprehensive studies of the nature of cognitive maps when he asked residents of three American cities (Boston, Los Angeles, and Jersey City) to draw maps of their city environments. He analyzed these drawings for commonalities in the features of people's mental images of their cities. This resulted in the identification of five major characteristics:

- (1) **Paths:** Major arteries of traffic flow through the city (e.g., Main Street)
- (2) **Edges:** Major lines (either natural or built) that divide areas of the city or delimit the boundaries (e.g., river)
- (3) **Districts:** Large sections of the city that have a distinct identity (e.g., "Chinatown");
- (4) **Nodes:** Points of intersection (cross)of major arteries (e.g., the corner of Twelfth Street and Vine)
- (5) **Landmarks:** Architecturally unique structures that can be seen from a distance and can be used as reference points (e.g., a tall building). Thus, the objective physical setting comes to be represented as "cognitive space," organized and structured mentally in terms of distinct "regions" of the environment.

According to the "anchor-point" hypothesis (the regionalization and hierarchical organization of cognitive space is brought about by the active role of salient cues in the environment. For example, primary nodes or other reference points "anchor" distinct regions in cognitive space. These components or reference points provide the "skeleton" of the individual's map. As we shall see, the degree and accuracy of the detail of the remainder of the map is a function of both aspects of the environment and individual differences.

Factors Affecting Cognitive Maps

Environmental Differences:

Environments differ from one another in the ease with which people can develop

cognitive maps of them. Lynch (1960) coined the term *legibility* to refer to the extent to which the spatial arrangement of a city facilitates a clear and unified image in the minds of its inhabitants. For example, Boston provides a clear center, the Boston Common, around which people organize their cognitive maps. On the other hand, Los Angeles does not appear to have any central core, but sprawls out in all directions, which inhibits an organized mental representation. Milgram and his associates (Milgram, Greenwald, Kessler, McKenna, & Waters, 1972) argued that recognizable areas of an environment are important for developing accurate cognitive maps.

They proposed the following formula for predicting the recognize ability of an area:

$R=f(C \times D)$. (It helps us to understand about those factors which helps us to understand the environment)

This formula is read, "The recognizability of an area (R) is determined by its centrality to population flow (C) and its architectural or social distinctive-ness (clearly) (D).". Thus, environments that have structures that stand out (such as a hilltop church) and are frequently passed by people facilitate a clear picture of the setting in the minds of the inhabitants.

Both environmental differences can be understood in terms of the "anchor-point" hypothesis in spatial cognition. That is, salient, objective physical cues in the environment facilitate the accurate organization of cognitive space, and the absence of such reference points inhibits accuracy.

Errors in Cognitive Maps:

The physical characteristics of a setting are not the only determinants of the accuracy and detail of cognitive maps. People are prone to a number of cognitive errors in the process of developing cognitive maps. Downs & Stea (1973) point out that most of us form *incomplete maps*, leaving out both minor and major details; we tend to *distort* our maps, by placing some areas closer together than they actually are and others farther apart than they actually are; and, we sometimes *augment* (zada krna) our maps by

including elements that do not actually exist. Further, we often give under prominence to areas of the environment that are personally meaningful or important to us. For example, Saarinen (1973) asked students from different countries to draw maps of the world. He found that the students tended to draw their own countries in the center of the map, drawing them larger than countries that are larger than their own. Other recent studies have demonstrated errors in estimations of the differences in elevation between locations (Garling, Book, Lindberg, & Arce, 1990) and memory for turns of varying angularity encountered during pathway traversal (Sadalla & Montello, 1989).

The above errors are most likely due to limitations in human spatial cognitive abilities, rather than to objective environmental characteristics. As with any kind of information-processing task, the accuracy of our spatial knowledge of the environment is unlikely to ever be complete, and some tasks are easier than others. For example, Teske & Balser (1986) asked individuals to identify various destinations in a city and their strategic ordering (i.e., plan itineraries), and Veitch & O'Connor (1987) asked students to do the same on their college campus. Subjects found planning itineraries more difficult because this requires a higher-level cognitive organization. That is, while the former task requires knowledge only of the route from point A to B, the latter also requires knowledge of the route from B to C and the

interrelationships among A, B, and C. Moeser (1988) has suggested such "survey maps" do not automatically develop in complex environments. She reported that student nurses failed to form survey maps of a hospital with a unique configuration, even after traversing it for two years. Further, these students performed worse on objective measures of cognitive mapping than did a control group of "naive" college students who were first asked to memorize the floor plans of the building.

Finally, Stanton (1986) investigated the relationship between "socio-spatial neighborhood" (the perception of a street network without continuous boundaries) and the experience of "home ground" (defined as the mental form and geographical extent of those places that evoke a feeling of being near home). She reported that only residents of city blocks that were no more than 460 feet long were able to think of their home ground as an experiential network, concluding that there may be a mental time limit to such cognitions. Thus, these studies all suggest that general cognitive limitations of information processing • are involved in spatial cognition as in any other type of complex task, and these limitations are a major source of errors in cognitive maps.

Individual Differences:

Investigators have also suggested that some people seem to be better at forming cognitive maps than others. For example, gender differences have been reported. Appleyard (1970) reported that the cognitive maps of men are generally more accurate than are those of women. More recently, Ward, Newcomb, & Overton (1986) examined how men and women gave directions from maps that had been memorized. Male subjects exhibited higher levels of cognitive organization, such as using more mileage estimates and cardinal directions (i.e., east, west, north, and south) and made fewer errors of commission or omission than did female subjects. Antes, McBride, & Collins (1988) reported that distance judgments of women were more affected by a change in travel paths through a city occasioned by the construction of a new connecting street than were men. They suggested that women based their judgments on inferences from travel paths, while men approached the task in a more spatial manner. Orleans & Schmidt (1972) reported that women's maps were more detailed for the home and neighborhood than were those of men, whereas men's cognitive maps were more comprehensive and complete for the larger surrounding environment. Finally, some investigators have reported socioeconomic differences, suggesting that the cognitive maps of people low in socioeconomic status are also less complete and accurate than are those individuals of higher socioeconomic status (Goodchild, 1974; Orleans, 1973).

Note that the individual differences listed above may not be due to differences in ability, but rather to differences in familiarity. That is, there is much evidence that people draw more detailed and accurate maps of areas with which they have had more experience (and thus are more familiar to them) than areas where they have spent little time (e.g., Appleyard, 1970; Evans, 1980; Holahan & Dobrowolny, 1978; Moore, 1974). Of course, it stands to reason that we would have better images of settings that are familiar to us than of unfamiliar places. Indeed, some of the studies on errors in cognitive maps discussed earlier also indicated that the extent of error can be moderated by experience.

Moreover, typically there are differences in mobility (نقل و حرکت) (among the groups

discussed above (i.e., in opportunity for travel through the setting). Thus, for example, if a husband works and the wife stay at home, it is not surprising that the husband would develop a better cognitive map of areas beyond the immediate neighbourhood while the wife would develop a more detailed map of the local environment. Similarly, people of higher socioeconomic status have much greater mobility to gain experience in the larger environment than do individuals of lower socioeconomic status. These suggestions are supported by the research of investigators who have controlled for mobility (e.g., Appleyard, 1976; Karan, Bladen, & Singh, 1980; Maurer & Baxter, 1972). While the individual differences reported above could be due to inherent differences in spatial ability, this seems unlikely to be the major reason. A study by Pearson and Jalongo (1986) measured spatial ability and environmental knowledge independently. Spatial ability accounted for only 14 percent of the variance in environmental knowledge. Thus, learning brought about by relevant experience in an environment is likely to be a more important determinant of the accuracy of cognitive maps than are the individual or cultural differences discussed above.

Lesson # 14

ENVIRONMENTAL EVALUATION

The question of how we come to evaluate an environment favorably or unfavorably is a complex one, yet it is an important one for predicting behavior. At a very general level, people prefer and approach environments they evaluate favorably and avoid environments they evaluate negatively. Environmental psychologists have dealt with the first question using a variety of operational definitions. For example, a favorable evaluation could be viewed as a preference for certain configurations of stimuli in some environments over that of other environments, a cognitive judgment of beauty, or as a positive affective reaction to the environment. Each of these approaches has led to the identification of different, though related, aspects of the physical environment as determinants of evaluative responses. The answer to the question "What aspects of the environment lead to a favourable evaluation?" is going to depend on the answer to the question "What is meant by a favourable evaluation?" Consideration of each of these approaches, as well as their implications for the relationship between environmental evaluation and behaviour follows.

Kaplan's Model of Environmental Preference®(everyone has their own preferences for the environment)

Would you rather be here or there? Kaplan (1975) developed a model for predicting preferences of some environments over others. This model provides a link between environmental cognition and evaluation, in that it assumes an important dimension of environmental preference to be the informational content of those environments. For example, Kaplan (1979) suggested that one basis for preferences is the ability of the individual to "make sense" out of the environment and the extent to which the environment involves the individual by motivating him or her to try to comprehend it.

Basic Assumptions of Kaplan's Model:

Individual base their preference for certain environments upon two factors:

- Can they make sense of it.
- Is there something in that environment that motivates them to learn more about that particular environment .

Kaplan & Kaplan (1978) identified four important factors influencing our preferences:

1- **Coherence:** ((بم آبنگی) chezun ki trteeb kya ha ? matching kya ha ??

The degree to which the environment is organized and harmonious, with clear patterns and relationships among its elements.

2- **Legibility:** ((معقولیت

The degree to which the environment is easy to understand and navigate, with distinctive features and landmarks that help orientation and wayfinding.

3- **Complexity:**

The degree to which the environment offers variety and diversity of elements, colors, shapes, and textures, without being overwhelming or chaotic.

4- **Mystery:** اسرار

The degree to which the environment invites exploration and discovery, with hidden or partially revealed elements that arouse curiosity and interest.

Impact of the Four Factors:

■ People prefer environments high on coherence and legibility to those that are low on these factors.

■ Greater degree of complexity and mystery produce a preference for a setting.

Kaplan (1987) has recently argued for a biological basis to such preferences.

Biological basis to Preferences:

There is a survival value to preferring environments that offer informational advantages over others.

The Kaplans' recent research points to the importance of mystery in predicting landscape preferences (Kaplan, Kaplan, & Brown, 1989), but others have applied their model to predict preferences for urban environments (Herzog, 1989) and interiors (Scott, 1989). **Environmental Aesthetics:**

Aesthetic is actually our sense to appreciate beauty. Is beauty in the eye of the beholder? Apart from the intuitive appeal of the notion that beauty is a subjective and relative concept, psychologists have attempted to identify objective dimensions of environments that lead to judgments of their aesthetic appeal. The assumption is that although there may indeed be individual differences in judgments of beauty, it is still possible to identify commonalities in what most people consider to be beautiful. For example, most people consider the Grand Canyon to be beautiful and a junk yard to be ugly. By researching a variety of settings, it is possible to identify dimensions along which different environments judged to be beautiful are similar.

We will discuss Environmental Aesthetics in detail in the next lecture

Lesson # 15

Environmental Evaluation (Affective Bases of Environmental Evaluation)

Environmental Aesthetics: (repeat)

Is beauty in the eye of the beholder? Apart from the intuitive appeal of the notion that beauty is a subjective and relative concept, psychologists have attempted to identify objective dimensions of environments that lead to judgments of their aesthetic appeal. The assumption is that although there may indeed be individual differences in judgments of beauty, it is still possible to identify commonalities in what most people consider to be beautiful. For example, most people consider the Grand Canyon to be beautiful and a junk yard to be ugly. By researching a variety of settings, it is possible to identify dimensions along which different environments judged to be beautiful are similar.

Berlyne (1960) conducted one of the first important series of studies along these lines. He identified four basic collative properties of environments.

Collative Properties of Environment:

Characteristics of the environment that cause us to compare present settings to previous settings we have encountered.

1. Complexity

It refers to the degree of variety in the elements of the environment

2. Novelty : is the extent to which stimuli not previously encountered or noticed are present. : وہ محرکات جس کا پہلے سامنا نہیں کیا گیا تھا اور نہ ہی دیکھا گیا تھا۔

3. Incongruity

Concerns the extent to which the environment contains stimuli that do not seem to "go together" harmoniously „ wife has 4 feet height and husband has 7 „

4. Surprisingness

It refers to the environment that contains elements that we do not expect to be present. Unexpected thing in environment

Berlyne Arguments 1974:

- Aesthetic judgments are most positive for environments at intermediate levels of each of these dimensions.
- we judge either too much or too little complexity, novelty, incongruity, or surprising of an environment.

Wohlwill (1976)

Wohlwill suggested that judgment of beauty increase consistently with increases in novelty and surprisingness, as well as with decreases in incongruity.

Collative properties of Environment and Exploratory Behaviours

There are two types of exploratory behaviours:

1. Specific Exploration

Specific exploration increases as the level of uncertainty in the environment increases. Presumably, uncertainty creates arousal, which leads to specific exploration to identify the source of the arousal.

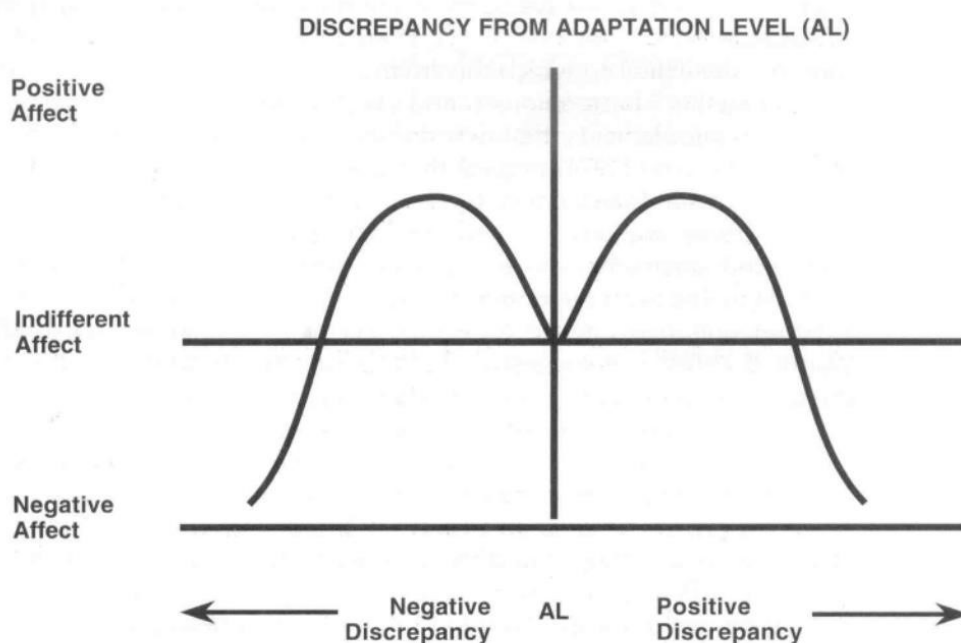
2. Diverive Exploration

Diverive exploration occurs when uncertainty-arousal levels are low. In other words, when the collative properties of an environment are at low levels, we feel under stimulated or bored, and we search the environment for ways to increase our arousal level.

Affective Bases for Environmental Evaluation:

Affective responses to environment are determined by degree of discrepancy ^{تضاد} between current levels of stimulation and the adaptation level (i.e. the level of stimulation, we have become accustomed to).

Wohlwill's Butterfly Curve Hypothesis:



Wohlwill proposed the butterfly curve hypothesis predicting a curvilinear relationship between positive affect and discrepancies resulting in either increases or decreases in arousal from adaptation level (see Figure above). That is, we view moderate increases or decreases in stimulation from adaption level as pleasant. However, extreme deviations in either Direction result in negative affect. This approach is useful in accounting for individual differences in evaluations of a setting. Individual differences in adaption level result from living in different environments, leading to what might be called the "one person's ceiling is another person's floor" effect. That is, the same environment might be perceived as overstimulation to one person while under stimulating to another. For example, a person raised in a small rural town in Iowa might perceive a city of 20,000 people overwhelming, while the same city might be perceived as boring to a person raised in Chicago. Note, however, that both individuals would experience negative affect, because the discrepancies from their adaptation levels are extreme, even though in different directions.

Mehrabian & Russell (1974)

The work of Mehrabian & Russell (1974) has added considerably to our understanding of the relationship among stimulation levels, arousal, and positive/negative affective responses to environments. Their model accounts for the relationship between these variables and behavior. Mehrabian and Russell propose the concept of **information rate** to define environmental stimulation level. Information rate refers to the average amount of information **impinging** **اثر انداز** on the senses per unit of time. This concept can be used to integrate the various dimensions of environmental stimulation discussed above, such as complexity, novelty, incongruity, surprisingness, mystery, and coherence, in that all of these dimensions contribute to the information rate of a setting. Mehrabian & Russell (1974) present a great deal of research demonstrating that **arousal** is a direct correlate of **information rate**, when information level is high automatically arousal will be high .and that approach behaviors in an environment (i.e., seeking out or desiring to remain in a setting) are greatest for intermediate levels of arousal. However, Mehrabian and Russell suggest that the curvilinear relationship between approach and arousal is moderated by the degree of pleasure a person experiences. Specifically, this relationship appears to hold only when pleasure is at an intermediate level. When pleasure is extremely high, approach behaviors strengthen with increases or decreases in arousal, and when pleasure is extremely low, both increases and decreases in arousal lead to avoidance behaviors.

We will discuss remaining part of Affective Bases for Environmental Evaluation in detail in the next lecture.

Lecture #16

Environmental Attitudes

Russell & Pratt Model of Emotional Reactions to Environment:

Russell & Pratt (1980), proposed a model of emotional reactions to environments in which arousal and pleasure are viewed as independent dimensions. This is illustrated in Figure below. As can be seen, the model depicts all possible combinations of pleasure and arousal. Thus, environments high in arousal can be perceived as pleasant (i.e./"exciting" settings, such as a football game), as can environments low in arousal (i.e./"relaxing" settings, as a picnic in the park). Note that we are likely to engage in approach behaviors to both types of environments. The same can be said of avoidance behaviors to unpleasant environments high in arousal.



FIGURE:

Circular ordering of eight terms to describe the emotional quality of environments (from Russell and Pratt, 1980).

("Distressing" settings, such as a final exam) and low in arousal ("gloomy" settings, such as a funeral). This circumplex model has been recently demonstrated to reliably differentiate between people's experiences in and preferences for suburban parks (Hull & Harvey, 1989).

Environmental Attitudes:

The relationships among the processes of environmental perception, cognition, and evaluation as well as their effects on behaviour can be best summarized in the notion of environmental attitudes. **Attitude:**

Attitude involves the way we think about, feel about and behave towards an object. Attitude consists of a cognitive component, affective component, and behavioural component.

- **Cognitive component:**

Processes by which we perceive our environment and how the perceptual processes are involved in developing our cognition and understanding of the environment.

- **Affective component:**

It refers to the emotional reactions or feelings an individual has towards an object, person, issue, or situation.

- **Behavioural component:**

Particular characteristics of cognitions that have implications **اثر رکھتی ہے** for one's overall attitude towards environment.

Environmental Attitude Formation:

Social psychologists have devoted considerable time and effort toward understanding the processes by which attitudes are formed. We will consider environmental attitude formation as a special case of the general process by which any attitude is formed. Most explanations of attitude development invoke learning principles such as classical conditioning, operant conditioning, and observational learning. Each of these principles will be discussed below, but **we will first consider two important cognitive foundations of attitudes: beliefs and values.**

Beliefs:

The way people feel towards things. Beliefs form the basis of our understanding and perception of the world.

Values:

Values are the characteristics patterns of behaving towards certain things. Values can include concepts like fairness, integrity, and respect for others, and they influence how people interact with the world around them

Daryl Bern (1970) has suggested that beliefs constitute the cognitive **"building blocks"** of attitudes.

Types of Beliefs

1. **Primitive beliefs**
2. **Higher order beliefs**

Primitive beliefs:

Primitive beliefs are non-conscious (i.e., they are accepted as givens, and are seldom consciously questioned). Primitive beliefs are either based on direct experience (e.g., the belief in the validity of our sense impressions) or on external authority (e.g., **the belief that if Mommy says so, it must be true**). The processes of sensation and perception discussed earlier are involved in the development of primitive beliefs. Because we have faith in our sensory impressions, if something smells or tastes bad, we will hold a negative attitude toward that thing. Alternatively, if Mommy tells us that we will get sick if we eat something, since we believe (at least as young children) that **"Mommy is always right and never lies,"** we will also develop a negative attitude toward that thing.

Higher order beliefs:

These beliefs involve the insertion of a conscious premise in the thought process of arriving at the belief, i.e. **the belief that air pollution is bad for health**. For example, if I believe that air pollution leads to respiratory illness, it is unlikely that I arrived at that belief through direct experience or because Mommy said so. Rather, it is more probable that I heard a newscast reporting the results of a study conducted by the office of the Surgeon General warning of the

health hazards of air pollution

We will discuss more about "Beliefs and its types" in next lecture.

Lecture # 17

Depth and Breadth of Beliefs

Depth:

Many premises leading to the same conclusion.

Breadth:

Many syllogisms/ deductive reasoning leading to the same conclusion.

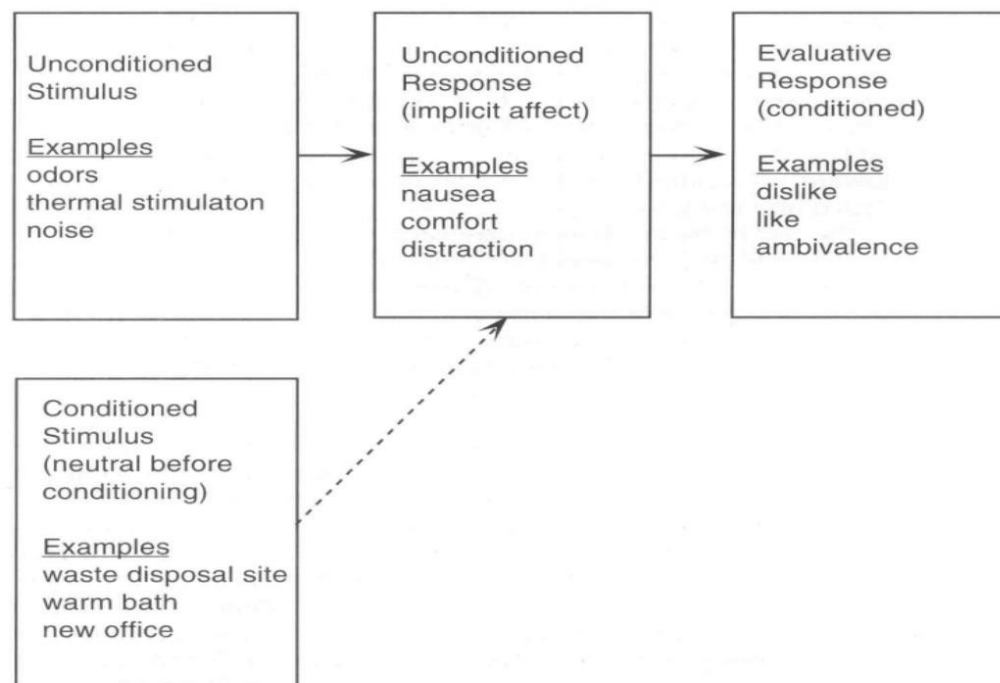
To complicate matters further, although the language of syllogisms, premises, and conclusions suggests that the process of arriving at higher-order beliefs is inherently logical and rational, this is not necessarily the case. Bern invokes the idea of psychological to account for irrational beliefs that are nonetheless internally consistent in the individual's cognitive structures. Further, it is possible to believe that something is bad for you but still evaluate the thing favorably. For example, I may know that smoking is bad for me, but simultaneously enjoy it (or alternatively, I may believe that exercise is good, but I may hate it anyway!).

Values:

Values can be defined as basic preferences for certain end states (see Rokeach, 1968). For example, equality is a value referring to an end state such as equal opportunity. Values serve as the functional basis for attitudes. That is, holding a particular attitude is a means for attaining a preferred end state. Thus, a person who strongly values equality is likely to be in favor of civil rights legislation. Similarly, if one values clean air, then that person is likely to evaluate actions to control pollution favorably. Indeed, Neuman (1986) has reported that values pertaining to environmental quality are positively related to beliefs about the efficacy and necessity of conservation and actual conservation behavior.

Affective Bases for Attitude:

Most social psychologists believe that the affective component of an attitude is learned. One important theory of attitude formation is The Byrne-Clore reinforcement-affect model (Byrne, 1971; Byrne & Clore, 1970). This model is based on principles of classical conditioning. Specifically, if an affectively "neutral" stimulus (called a conditioned stimulus) is paired with a stimulus that does elicit an effective response (called an unconditioned stimulus), and then the previously neutral stimulus will acquire the same ability to produce the response. The affective response to the unconditioned stimulus is referred to as an unconditioned response, because it occurs unconditionally (i.e., without the organism having to learn the response). The acquired response is called a conditioned response, because the organism must be conditioned to make the response in the presence of that stimulus (i.e., the organism must "learn" the response through association). This model can be readily applied to the conditioning of the affective component of environmental attitudes (see Figure below).



Do not think about the environment if the stimulation levels are within some optimal range). However, if some quality of the environment changes to deviate from the optimal stimulation level, a negative affective response automatically occurs. Through association, the entire environment comes to elicit the same negative affect. For example, when you first encountered your dormitory room, your affective response to the desk in the room (the "conditioned stimulus") was likely to have been neutral. However, if when you sat down to study at the desk, your neighbor turned rock music (the "unconditioned stimulus") on his or her stereo at full volume; you may have experienced negative affect (the "unconditioned response"). If this happened repeatedly, you would develop a negative affective response to your desk (the "conditioned response").

Behavioural Bases of Attitudes

Operant Conditioning (Skinner, 1938)

Any behavior that is followed by a pleasant consequence will increase in frequency. Behaviors that lead to unpleasant consequences will become extinguished, بُجھ جانا that is, they will not be repeated. The expression of a particular attitude is a form of behavior, and can therefore be operantly conditioned as any other behavior. Thus, we learn to express attitudes that lead to favorable consequences for us, and we avoid attitudes that pose aversive نفرت انگیز consequences. For example, he used a specially designed box, known as the Skinner Box, to analyze the learning behavior's of pigeons. In these experiments, pigeons were placed in the box and had to peck at a disc to receive food as a reward. Skinner manipulated various conditions to study the effects of different reinforcement schedules on the pigeons' behavior.

We will discuss more about "Instrumental conditioning" in next lecture.

Lecture #18

Operant Conditioning (Skinner, 1938)

Operant conditioning is a learning theory developed by B.F. Skinner that describes how behaviors are influenced by their consequences. It's based on the idea that behaviors followed by positive outcomes are more likely to be repeated, while those followed by negative outcomes are less likely to be repeated. Here's a detailed explanation:

How Operant Conditioning Works:

- **Reinforcement:**

This is any consequence that strengthens the behavior it follows یہ کوئی بھی نتیجہ ہے جو اس کی پیروی کرنے والے رویے کو مضبوط کرتا ہے۔ There are two types of reinforcement:

- **Positive Reinforcement:**

Adding a desirable stimulus to increase a behavior. For example, giving a child a treat when they clean their room.

- **Negative Reinforcement:**

Removing an aversive stimulus to increase a behavior. For example, buckling the seatbelt to stop the car from beeping.

- **Punishment:**

This is any consequence that decreases the behavior it follows. Like reinforcement, punishment can also be positive or negative:

- **Positive Punishment:**

- Adding an aversive stimulus to decrease a behavior. For example, scolding a pet for misbehaving

- **Negative Punishment:**

Removing a desirable stimulus to decrease a behavior. For example, taking away a teenager's gaming privileges for not doing homework.

Social Bases of Attitudes

Another important way in which attitudes are formed is through observation of other people's expression of attitudes and the consequences of their attitudes.

Social Learning:

The process of learning via observation of a model's behavior is known as social learning.

Vicarious Learning:

The means by which we are influenced by the consequences of the model's behavior is known as vicarious conditioning. (agr log logun hmary behavior py positive response dyn to ,hmara whi behave repeat hoga ,, or agr negative response hoga to hm us behave sy avoid kren gy ,, yhi previous mn b likha hua va ha ,, whi sy krna ha)

If an individual observes another person expressing a particular attitude, and also observes that the attitude led to favorable consequences, then the individual will imitate the model's behavior in anticipation of incurring the same favorable consequences. For example, if a person who works for an oil company observes a co-worker express opposition to a windfall profits tax, and the co-worker is praised by the boss, then the individual is likely to imitate this opposition in anticipation of also gaining the boss's approval.

Another way in which social influences can impact attitudes is through the dynamics of interpersonal processes. Put simply, if my friends have pro-environmental attitudes, and I enjoy my friends' company, then I am likely to adopt similar attitudes. Manzo & Weinstein (1987) studied differences in active and nonactive members of the Sierra Club and reported significant differences in commitment to environmental protection as a function of club-related friendships. Thus, friends tend mutually to reinforce attitudes toward the environment.

Attitudes and Behavior

A major assumption of research on attitude formation is that attitudes mediate behaviors which are consistent with those attitudes. Thus, if I hold an attitude favoring energy conservation, it would be expected that this attitude would mediate behaviors such as walking instead of driving to work, or turning my thermostat down during the winter and up during the summer. Although this may seem like an obvious and reasonable assumption, in recent year's research has led social psychologists to question the degree to which attitudes do, indeed, reliably predict behavior (e.g., Wicker, 1969). These studies have suggested that there is much less consistency between measured attitudes and subsequent behaviors than had previously been assumed.

Attitude-Behavior Consistency

The major principle of consistency theories of attitude-behavior relationships (see Festinger, 1957) is that people strive to maintain logical consistency between cognitions about their attitudes and cognitions about relevant behaviors in which they engage. Thus, if a person has the opinion that air pollution should be brought under control, a logically consistent behavior would be to vote for measures requiring utility companies to install pollution-control equipment in energy plants.

Cognitive Discomfort:

If you dislike someone it is your attitude and if you deal with him negatively then it is your behavior. When you dislike someone and yet you make a good deal due to some compulsion, you become uncomfortable. this state called discomfort .

اگر آپ کسی کو ڈس لائیک کرتے ہیں یہ آپ کا ایٹیٹیوڈ ہے اور آپ اس کے ساتھ نیگیٹیوولی ڈیل کرتے ہیں تو یہ آپ کا بیہیو ہے جب آپ کسی کو ڈس لائیک کرتے ہیں پھر بھی کسی مجبوری کی وجہ سے آپ اچھا ڈیل کرتے ہیں تو آپ ڈس کمفرٹ ہو جاتے ہیں

Cognitive discomfort, often referred to as cognitive dissonance, is the mental stress or unease experienced when an individual holds two or more contradictory beliefs, values, or attitudes simultaneously. This state of tension arises when there is an inconsistency between what a person believes and how they behave.

In context to the above-mentioned example of air pollution, If this person knows that he or she voted against such a measure, the perceived attitude-behavior inconsistency would create a state of cognitive discomfort, which would motivate the individual to attempt to restore consistency. This would be accomplished either by changing the attitude (e.g., deciding that pollution control is not as important as controlling the costs of energy production), or by

changing the behavior (e.g., by voting in favor of the next such measure).